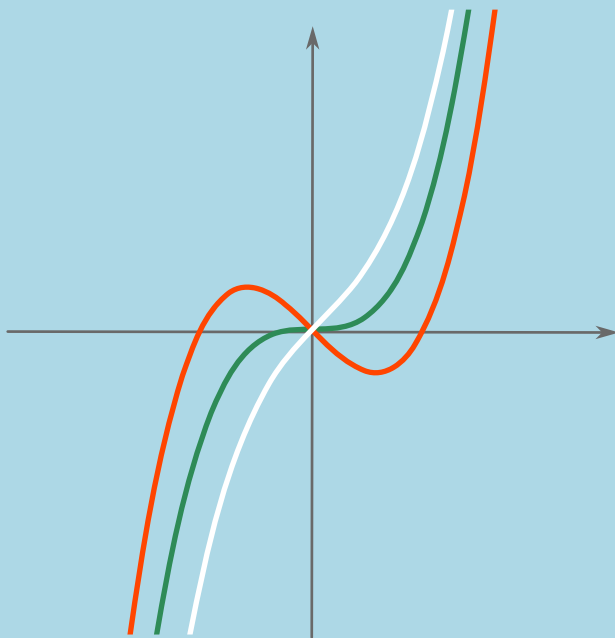


Lev Pontryagin

MATHEMATICAL ANALYSIS

For School Students

(A Book for School Students... And Not Only for Them!)



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MATHEMATICAL ANALYSIS

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Translated from the Russian by
Dimitr Mazanav

The Mir Titles Project 2025

Lev Semenovich Pontryagin

Mathematical Analysis for School Students

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ANNOTATION

This book is intended as an introduction to mathematical analysis. It includes material covering all sections of mathematical analysis taught at the secondary school level. The book examines derivatives of polynomials, trigonometric functions, exponential and logarithmic functions. The integral is presented as an operation inverse to differentiation, as the area under a graph, and as the limit of finite sums. At the end of the book, exercises are provided for each section.

The emphasis is placed not on the rigour of exposition, but on computational technique.

For senior school students and for anyone beginning to study mathematical analysis.

Preface

This small book, about five printers sheets in length, is intended — with some luck — to become a textbook of mathematical analysis for secondary schools. It contains everything that could be included in any version of a school curriculum.

The book does not begin with the definition of a limit or the rules for calculating it. The limit is treated as something self-evident and is introduced through the definitions of the tangent and the derivative. That is where the book begins.

Then it proceeds to compute the derivatives of polynomials and trigonometric functions, and introduces the rules for differentiating products, quotients, and composite functions. In between, Rolles Theorem and

Lagranges Formula are proved. On this basis, functions are studied: intervals of increase and decrease are identified, as well as maxima and minima.

The integral is presented in three forms: as the operation inverse to differentiation, as the area under a graph, and as the limit of finite sums. After this, the function e^x is studied carefully as the limit of a sequence of polynomials $(1 + \frac{1}{n})^n$ as the integer n tends to infinity. The derivatives of the functions e^x and $\ln x$ are calculated.

At the end, each paragraph is followed by exercises — not many, but some of them are rather challenging. The emphasis of the book is not on logical rigour, but on computational technique. As a popular-level book, it can serve as a first introduction to mathematical analysis.

Since I myself have never taught in a secondary school, in writing this book I relied on the common sense of a qualified mathematician and on my personal recollections of how I once perceived these ideas.

During my school years, analysis was not part of the curriculum. Nevertheless, even before entering university, I was quite familiar with it. I knew what a derivative and an integral were, and I could use this

apparatus to solve problems. At the same time, I had not the slightest idea about the theory of limits. I only learned about its existence at university, and I was extremely surprised by it.

I believe that it is not advisable to begin the study of analysis in secondary school with the theory of limits. One must remember that the theory of limits historically arose as a superstructure on top of already existing analysis. A thorough study of such concepts as limits and continuous functions may lead to boredom or even aversion. I remember, as a schoolboy, reading a proof in some course of analysis about the theorem stating that a continuous function takes all intermediate values. This reading caused me great bewilderment and irritation at the time.

A sensible person should perceive the graph of a function as the well-finished edge of a smooth metal plate. With such a perception of a graph, a tangent on its convex part should be seen as the edge of a ruler tightly pressed against the convex part of the plate's edge. Therefore, neither the existence of a tangent nor the existence of a derivative should raise any doubt. In the same way, there should be no doubt that the area of such a plate exists, and hence there is no doubt in the existence of an integral.

I would like a school student, when studying geometry, to perceive a triangle as if it were made of a thin metal plate, so that it can be held in ones hands, moved to another place, and turned inside out. This does not mean that such should be the definition of a triangle, but I think the perception of it should be exactly that.

Based on such methodological considerations, I begin the presentation of analysis not with the definition of a limit, but with the definition of a tangent and a derivative.

It seems to me that only the material covered in Sections 1 to 7 should be included in the secondary school curriculum. The rigorous description of the function e^x , which is the focus of Sections 8 to 10, appears to me to be extremely difficult. Nevertheless, I include it in view of the curriculum requirements.

In the same way, in order to meet the requirements of the syllabus, I provide some information about limits and continuous functions, but only in the epilogue (§ 13).

In conclusion, I would like to express my gratitude to V.R. Telesnin for the great assistance he gave me in writing and editing this book.

Contents

Preface	v
1 The Derivative	1
The Derivative and the Tangent	2
Continuity of a Function	13
2 Computing the Derivative of a Polynomial	15
3 Maximum and Minimum	23
Positivity and Negativity of the Derivative . .	24
Maximum and Minimum	26
Rolle's Theorem	28
Lagrange's Formula of Finite Increments of a Function	30
The Second Derivative	33

Contents

Distinguishing Between a Maximum and a Minimum	34
4 Study Of Functions	36
Inflection Point	42
Exercises	46
5 Derivatives of Trigonometric Functions	49
Derivative of a Product and a Quotient	53
The Derivative of $\tan x$ function	55
Derivative of a Composite Function	56
Inverse Function	58
Derivative of a Rational Power of x	62
6 Indefinite Integral	65
7 Definite Integral	75
Definite Integral as the Limit of a Sequence of Finite Sums	82
8 Postulate of Convergence	85
Postulate of Convergence	88
9 Newton's Binomial and the Sum of a Geometric Progression	92
Newton's Binomial	92
Sum of a Geometric Progression	96

Contents

10	The Function e^x	98
	Study of the Function $\omega_n(x)$	99
	Case $ x \leq 1$	102
	Fundamental Property of the Function $\exp(x)$	105
	The Number e and the Function $\exp x$	108
	The Derivative of the Function e^x	110
11	The Function $\ln x$	112
12	Expansion of the Function e^x into a Series	114
13	Epilogue: On the Theory of Limits	117
	The Theory of Limits	118
	Continuous Functions	122
14	Exercises	124
	Section 1	124
	Section 2	128
	Section 3	128
	Section 4	130
	Section 5	137
	Section 6	145
	Section 7	151
	Section 8	154
	Section 10	155
	Section 11	156
	Section 13	159

1 The Derivative

When studying a function, one of the most important concepts is its derivative. If a certain function

$$y = f(x) \tag{1.1}$$

is given, one can compute the function $f'(x)$, which is called the *derivative* of the function $f(x)$. The quantity $f'(x)$ describes the rate of change of the quantity y with respect to the change in the quantity x . This is, of course, not a formal definition of the derivative, but rather an intuitive description. Let us reinforce this idea by considering one particular case. If the relation

$$y = kx$$

is a proportional one, then the rate of change of y with respect to x is naturally k , i.e., in this case we must have $f'(x) = k$.

In this case, the derivative has a clear mechanical meaning. If we interpret x as time and y as the distance covered during this time, then k is the speed of the moving point. Here, the rate $f'(x)$ of change of y with respect to x is a constant value, but in the case of a more complex dependence of y on x , the derivative $f'(x)$ itself becomes a function of the variable x .

The derivative is used in the study of physical processes, where various physical quantities change over time, and the rate of their change plays a crucial role. We shall, however, begin with a geometric application of the derivative and use this example to clarify the very concept of a derivative.

The Derivative and the Tangent

Let us construct the graph of the function $f(x)$ (see (1.1)) in a Cartesian coordinate system. As usual, we draw the horizontal axis of abscissas on the plane of our diagram, choose the direction from left to right, and draw the vertical axis of ordinates, choosing the direction from bottom to top (see Figure 1). The graph of the function $f(x)$ in this coordinate system is a curve, which we shall denote by L .

Let us set the task of giving a logical definition of

the tangent to the curve L at a point A on this curve and of computing the quantities that define this tangent.

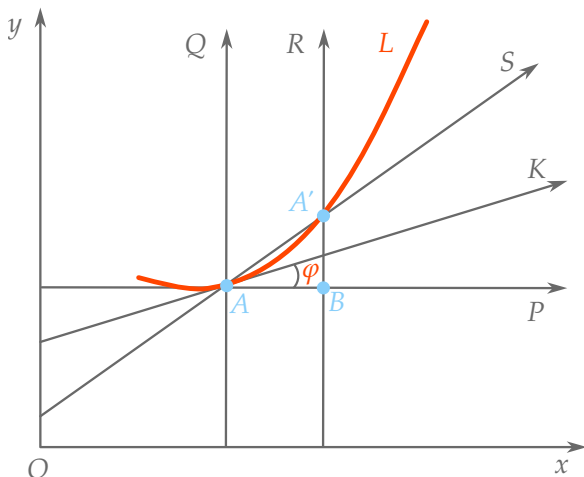


Figure 1: Finding the tangent to a curve L at a given point A .

To provide a logical definition of the tangent, let us choose another point A' near the point A , which we consider fixed for now. The point A' is close to A but distinct from it and is allowed to move. Through the points A and A' , we draw a straight line S , called a *secant line* to L , since it intersects L at the two points A and A' . The point A' , which lies on the curve L , may be either to the right or to the left of the point A .

Now let us begin to move the point A' along the curve L , bringing it arbitrarily close to the point A . If, during this motion of the point A' , the secant S , passing through the fixed point A and the moving point A' , rotates and tends toward a limiting position, a straight line K passing through point A , then this line K is called the *tangent* to the curve L at point A .

It is required that the line K does not depend on the side from which the point A' approaches point A . If the secant line S does not approach any definite position K , then the curve L is said to have no tangent at point A .

The tangent K passes through a fixed point A . If the line K is not vertical, then to compute the angle φ of its inclination to the horizontal axis, it suffices to compute $\tan \varphi$.

To compute the value of $\tan \varphi$, we shall first compute $\tan \psi$, the tangent of the inclination angle ψ of the secant line S with respect to the x -axis.

Let us denote the abscissa and ordinate of point a by x and y respectively:

$$a = (x, y) \tag{1.2}$$

where x and y are related by equation (1.1), since point A lies on the graph L of the function $f(x)$.

Similarly, let us denote by ξ and η the abscissa and ordinate of point A' :

$$\bar{a} = (\xi, \eta) \quad (1.3)$$

where ξ and η are related by the equation

$$\eta = f(\xi) \quad (1.4)$$

since point A' also lies on the graph L of the function $f(x)$.

Now, let us draw two straight lines through point A : a horizontal line P and a vertical line Q . These lines are parallel to the x -axis and y -axis, respectively, of the original coordinate system. We shall choose their directions to correspond with the directions previously selected for the coordinate axes. The lines P and Q , considered as the new x -axis and y -axis, respectively, define a new coordinate system in the plane with its origin at point A .

Since the secant line S cannot be vertical, we can clearly define a direction along it from left to right. Because the new x -axis P is parallel to the original x -axis, to find the angle ψ , it suffices to compute the angle between the positive direction on line P and the positive direction on the secant S .

This angle ψ is less than a right angle, but it can be either positive or negative. The new coordinate

system with axes P and Q divides the plane into four quadrants. The angle ψ is positive if the secant line S moves from the third quadrant to the first, and it is negative if the secant line S moves from the second quadrant to the fourth.

The abscissa and ordinate of point A' in the new coordinate system are respectively equal to the quantities:

$$(\xi - x), \quad (\eta - y). \quad (1.5)$$

To compute the angle ψ , let us draw a vertical line R through point A' , and denote by B the point of intersection of line R with line P . We now consider the right-angled triangle $A'BA$, where B is the vertex of the right angle.

Disregarding the sign of the angle ψ , its magnitude is equal to the angle $BA'A$ at vertex A' of the triangle. The tangent of this angle equals the length of the leg $\overline{BA'}$ divided by the length of the leg $\overline{A'A}$. Thus, we have the formula:

$$|\tan \psi| = \frac{\overline{BA'}}{\overline{AB}}. \quad (1.6)$$

The length $\overline{BA'}$ is the absolute value of the ordinate of point A in the new coordinate system, i.e., equal to $|\eta - y|$ (see (1.5)). Similarly, the length $\overline{AB}$ is the

absolute value of the abscissa of point A' in the new coordinate system, i.e., equal to $|\xi - x|$.

Thus, from equation (1.6), it follows that:

$$|\tan \psi| = \frac{|\eta - y|}{|\xi - x|}. \quad (1.7)$$

We shall now prove that:

$$\tan \psi = \frac{\eta - y}{\xi - x} \quad (1.8)$$

Note that if point A lies in the first or third quadrant, then its abscissa and ordinate have the same sign, and hence the right-hand side of equation (1.8) is positive.

In this case, the quantity $\tan \psi$ is also positive, as the angle ψ is positive. If the point A' lies in the second or fourth quadrant, then its abscissa and ordinate (see (1.5)) have opposite signs, and hence the right-hand side of equation (1.8) is negative. In this case, $\tan \psi$ is also negative, since the angle ψ is negative.

Thus, formula (1.8) is proven. Now, substituting the values of y and η from equations (1.1) and (1.4), we obtain:

$$\tan \psi = \frac{f(\xi) - f(x)}{\xi - x}. \quad (1.9)$$

When point A approaches point A without bound, its abscissa ξ approaches the abscissa x of point A without

bound. This is written as:

$$\eta \rightarrow x. \quad (1.10)$$

To find the value of $\tan \varphi$, we need to compute what $\tan \psi$ tends to as $\xi \rightarrow x$. This can be written as:

$$\tan \psi \rightarrow \tan \varphi \quad \text{as } \xi \rightarrow x. \quad (1.11)$$

In higher mathematics, this relation, consisting of two expressions, is written as a single equation:

$$\lim_{\xi \rightarrow x} \tan \psi = \tan \varphi. \quad (1.12)$$

Or, substituting $\tan \psi$ from equation (1.9), this last equality becomes:

$$\tan \varphi = \lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x} \quad (1.13)$$

The symbol $\lim$ is an abbreviation of the Latin word *limit*.

To rigorously describe the operation in equation (1.13) performed on the fraction

$$\frac{f(\xi) - f(x)}{\xi - x} \quad (1.14)$$

we would need to precisely define what the symbol $\rightarrow$ means — that is, explain the *concept of a variable tending*

towards a constant. But here, we rely on the intuitive understanding of this process.

Note that in expression (1.14), we cannot simply substitute $\xi = x$, as this would result in a fraction with both numerator and denominator equal to zero. Hence, it is necessary to consider the *process* of ξ approaching the constant x and observe the behaviour of expression (1.14) during this process.

To explain the concept of a limiting process using a simple example, consider the case where the function $f(x)$ is given by the formula:

$$y = f(x) = x^2. \quad (1.15)$$

In this case, the fraction from equation (1.14) becomes:

$$\frac{f(\xi) - f(x)}{\xi - x} = \frac{\xi^2 - x^2}{\xi - x} = \xi + x. \quad (1.16)$$

In the right-hand side of this equation, we can already replace the variable ξ with x without obtaining the indeterminate form $0/0$. Therefore, in this particular case, we have:

$$\lim_{\xi \rightarrow x} \frac{\xi^2 - x^2}{\xi - x} = \lim_{\xi \rightarrow x} (\xi + x) = 2x. \quad (1.17)$$

Thus, we have established that:

$$\frac{\xi^2 - x^2}{\xi - x} \rightarrow 2x \quad \text{as } \xi \rightarrow x, \quad (1.18)$$

or, equivalently:

$$\lim_{\xi \rightarrow x} \frac{\xi^2 - x^2}{\xi - x} = 2x. \quad (1.19)$$

The quantity

$$\lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x} \quad (1.20)$$

is called the *derivative* of an arbitrary function $f(x)$ at the point x , and it is denoted by $f'(x)$. By definition, we thus have:

$$f'(x) = \lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x} = \tan \varphi \quad (1.21)$$

(see equation (1.13)). It may happen, however, that as $\xi \rightarrow x$, the quantity in equation (1.14) *does not* approach *any* limit. In this case, it is said that the derivative of the function $f(x)$ at the point x *does not exist*.

Thus, formula (1.19) shows that for the function in (1.15) we have

$$f'(x) = 2x. \quad (1.22)$$

It should be noted that in formula (1.21) we do not separately consider the two cases when ξ approaches

x from the right side and when ξ approaches from the left side. In the first case, ξ approaches x from values greater than x (i.e., decreasing), and in the second case, ξ approaches x from values less than x (i.e., increasing). In both of these approaches, the result of calculating the derivative *must* be the same. Only then is it said that the derivative exists at the point x .

If the line K is vertical, then $\tan \psi$ increases without bound as $\xi \rightarrow x$. Thus, the expression (1.14) in this case increases without bound and therefore *does not tend to any limit*, so the derivative at this point does not exist. However, it is sometimes said, as a convention, that the derivative at such a point becomes infinite. From what has been said, it follows that the derivative of the function $f(x)$ exists if and only if the tangent K exists at point A to the curve L , and this tangent is not vertical.

However, it is easy to indicate a function for which the left-hand and right-hand limits of expression (1.14) yield different results. Consider, for example, the function $f(x)$ defined by the equation:

$$y = f(x) = absx + x^2. \quad (1.23)$$

We shall compute the derivative of this function $f(x)$ at $x = 0$. Thus, in this case we have:

$$\frac{f(\xi) - f(0)}{\xi - 0} = \frac{|\xi| + \xi^2}{\xi} = \frac{|\xi|}{\xi} + \xi. \quad (1.24)$$

When ξ is positive, $|\xi| = \xi$, and when ξ is negative, $|\xi| = -\xi$. Thus, we have:

$$\frac{|\xi|}{\xi} = +1 \quad \text{when } \xi > 0, \quad (1.25)$$

and

$$\frac{|\xi|}{\xi} = -1 \quad \text{when } \xi < 0. \quad (1.26)$$

Therefore,

$$\lim_{\xi \rightarrow 0} \frac{|\xi| + \xi^2}{\xi} = +1, \quad \text{when } \xi > 0, \quad (1.27)$$

$$\lim_{\xi \rightarrow 0} \frac{|\xi| + \xi^2}{\xi} = -1, \quad \text{when } \xi < 0. \quad (1.28)$$

It follows that the limit in expression (1.24) depends on the direction from which ξ approaches 0: from the right or from the left. In this case, it is said that the function $f(x)$ (see (1.23)) does not have a derivative at $x = 0$, and the corresponding graph does not have a tangent line there.

There are also more complicated cases where formula (1.21) does not determine the value of $f'(x)$.

The operation of finding the derivative of a function $f(x)$ (see (1.21)) is often called the *differentiation* of the function $f(x)$, so that a function which has a derivative

at a point x is said to be *differentiable* at that point. All functions we consider henceforth will be differentiable at all points, except perhaps at certain individual points where the derivative becomes infinite.

In the future, we shall assume that any function under consideration is differentiable at all points, without explicitly stating this. Only in cases where the derivative becomes infinite at specific points will we make special mention of this fact.

Continuity of a Function

Let us rewrite formula (1.21) in a slightly different way. Let us set

$$k = k(\xi) = \frac{f(\xi) - f(x)}{\xi - x}. \quad (1.29)$$

Then,

$$f'(x) = \lim_{\xi \rightarrow x} k(\xi). \quad (1.30)$$

The quantity $k(x)$ is not defined by formula (1.29). Let us define it by setting $k(x) = f'(x)$. We then have

$$f(\xi) - f(x) = k(\xi) \cdot k(\xi - x), \quad (1.31)$$

where $k(\xi)$ satisfies condition (1.30). Formula (1.31) also holds when $\xi = x$.

The difference $\xi - x$ is called the *increment of the argument*, and the difference $f(\xi) - f(x)$ is called the *increment of the function*. Formula (1.31) shows that the increment of the function is proportional to the increment of the argument, with the proportionality coefficient $k(\xi)$ being approximately equal to $f'(x)$.

Passing to the limit in equation (1.31) as $\xi \rightarrow x$, we obtain:

$$\lim_{\xi \rightarrow x} (f(\xi) - f(x)) = \lim_{\xi \rightarrow x} k(\xi) \cdot \lim_{\xi \rightarrow x} (\xi - x) = f'(x) \cdot 0 = 0,$$

whence it follows that

$$\lim_{\xi \rightarrow x} f(\xi) = f(x). \quad (1.32)$$

If the function $f(x)$ satisfies condition (1.32) at a given x , then it is said to be *continuous* at that value of x . If a function is continuous for all values of the argument, it is called a *continuous function*.

Thus, we have shown that if the function $f(x)$ has a derivative at a given point x , then it is continuous at that point. Moreover, if a function is differentiable, i.e., differentiable at all values of x , then it is continuous.

2 Computing the Derivative of a Polynomial

Here we compute the derivative of the function

$$y = f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n, \quad (2.1)$$

i.e., the derivative of an arbitrary polynomial with respect to x with constant coefficients $a_0, a_1, \dots, a_{n-1}, a_n$.

It will sometimes be convenient for us to use a slightly modified notation for the derivative $f'(x)$ of the function $f(x)$. Specifically, we shall sometimes denote this derivative as $(f(x))'$, that is, we set

$$f'(x) = (f(x))'. \quad (2.2)$$

Using this notation, we may now write the two formu-

2 Computing the Derivative of a Polynomial

las (1.15) and (1.22) from Section 1 as a single formula:

$$(x^2)' = 2x.$$

We shall first compute the derivative of the simplest polynomial of degree n , which reduces to a monomial. That is,

$$y = f(x) = x^n. \quad (2.3)$$

To compute the derivative of the function given in equation (2.3), we shall make use of a very simple, yet very important formula from algebra, the proof of which will be provided here.

To state and prove this algebraic formula, we introduce the polynomial $\varphi_k(u, v)$ in two variables u and v , defined by the formula

$$\varphi_k(u, v) = u^k + u^{k-2}v + \dots + uv^{k-1} + v^k \quad (2.4)$$

Thus, the polynomial $\varphi_k(u, v)$ represents the sum of all monomials of the form $u^i v^j$ where i and j are non-negative integers satisfying the condition $i + j = k$.

Multiplying the polynomial $\varphi_k(u, v)$ by the quantity u , we form the polynomial

$$\varphi_k(u, v) \cdot u. \quad (2.5)$$

This polynomial represents the sum of all monomials of the form $u^{i+1}v^j$, where i and j are non-negative

2 Computing the Derivative of a Polynomial

integers related by $i + j = k$. Here is the translation and Thus, the polynomial (2.5) represents the sum of all monomials of the form $u^p v^q$, where p and q are non-negative integers satisfying the conditions

$$p \geq 1, \quad p + q = k + 1.$$

Thus, polynomial (2.5) contains all terms included in the polynomial $\varphi_{k+1}(u, v)$, except for the single term v^{k+1} . Therefore, we have the equality:

$$\varphi_k(u, v) \cdot u = \varphi_{k+1}(u, v) - v^{k+1}. \quad (2.6)$$

Similarly, by multiplying the polynomial $\varphi_k(u, v)$ by v , we obtain the formula:

$$\varphi_k(u, v) \cdot v = \varphi_{k+1}(u, v) - u^{k+1}. \quad (2.7)$$

Subtracting equation (2.7) from equation (2.6), we get:

$$\varphi_k(u, v)(u - v) = u^{k+1} - v^{k+1}. \quad (2.8)$$

Replacing $k + 1$ with n in the identity (2.8) and dividing the resulting expression by $u - v$, we obtain the following important formula:

$$\frac{u^n - v^n}{u - v} = \varphi_{n-1}(u, v), \quad (2.9)$$

where $\varphi_{n-1}(u, v)$ is defined by the expression

$$\varphi_{n-1}(u, v) = u^{n-1} + u^{n-2}v + \dots + uv^{n-2} + v^{n-1}. \quad (2.10)$$

2 Computing the Derivative of a Polynomial

It should be emphasised that here $n \geq 1$, since $n = k + 1$ with $k \geq 0$. Note that the polynomial $\varphi_{n-1}(u, v)$ contains exactly n terms.

Using formula (2.9), we can easily compute the derivative of the function x^n . According to the rules stated in Section 1 (see equation (1.21)), we must first form the difference quotient

$$\frac{\xi^n - x^n}{\xi - x} \quad (2.11)$$

and then take the limit as $\xi \rightarrow x$. By the algebraic identity (2.9), we have

$$\frac{\xi^n - x^n}{\xi - x} = \xi^{n-1} + \xi^{n-2}x + \dots + \xi x^{n-2} + x^{n-1}, \quad (2.12)$$

where the right-hand side consists of exactly n terms. When passing to the limit as $\xi \rightarrow x$, we must replace ξ with x on the right side of equality (2.12). In doing so, each term $\xi^i x^j$, where $i + j = n - 1$, will turn into the term x^{n-1} . Thus, we obtain

$$(x^n)' = \lim_{\xi \rightarrow x} \frac{\xi^n - x^n}{\xi - x} = nx^{n-1},$$

and finally, we have

$$(x^n)' = nx^{n-1}. \quad (2.13)$$

2 Computing the Derivative of a Polynomial

In the proof of formula (2.13), we could not consider the case $n = 0$, since formula (2.9) is valid only for $n \geq 1$. Thus, we have not yet computed the derivative of the function $x^0 = 1$. It is simpler to compute the derivative of the function $f(x) = c$, where c is a constant. For this function, we have:

$$\frac{f(\xi) - f(x)}{\xi - x} = \frac{c - c}{\xi - x} = 0.$$

Thus, we have

$$c' = 0, \quad (2.14)$$

that is, the *derivative of a constant is zero*.

To pass from the simplest monomial x^n to the general polynomial (2.1), we need to establish two general rules for finding derivatives: the derivative of a sum of two functions, and the derivative of the product of a constant with a function. These rules are as follows. If $f(x)$ and $g(x)$ are two functions, then we have

$$(f(x) + g(x))' = f'(x) + g'(x). \quad (2.15)$$

In words: *the derivative of the sum of two functions is equal to the sum of the derivatives of the terms*.

Further, if c is a constant and $f(x)$ is some function, then we have

$$(cf(x))' = cf'(x). \quad (2.16)$$

2 Computing the Derivative of a Polynomial

In words: *the derivative of the product of a constant and a function is equal to the product of the constant and the derivative of the function.*

Let us first prove rule (2.15). We have:

$$\begin{aligned}(f(x) + g(x))' &= \lim_{\xi \rightarrow x} \frac{(f(\xi) + g(\xi)) - (f(x) + g(x))}{\xi - x} \\&= \lim_{\xi \rightarrow x} \left[\frac{f(\xi) - f(x)}{\xi - x} + \frac{g(\xi) - g(x)}{\xi - x} \right] \\&= \lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x} + \lim_{\xi \rightarrow x} \frac{g(\xi) - g(x)}{\xi - x} \\&= f'(x) + g'(x).\end{aligned}$$

Thus, rule (2.15) is proven. Similarly, rule (2.16) is proven as follows. We have:

$$\begin{aligned}(cf(x))' &= \lim_{\xi \rightarrow x} \frac{cf(\xi) - cf(x)}{\xi - x} \\&= \lim_{\xi \rightarrow x} c \cdot \frac{f(\xi) - f(x)}{\xi - x} \\&= c \lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x} \\&= cf'(x)\end{aligned}$$

Therefore, rule (2.16) is also proven.

From rules (2.15) and (2.16), we derive a generalised rule. Suppose we are given functions $f_1(x), f_2(x), \dots, f_m(x)$

2 Computing the Derivative of a Polynomial

and constants $c_1, c_2, \dots, c_m$. Then we have the following rule:

$$\begin{aligned} & (c_1 f_1(x) + c_2 f_2(x) + \dots + c_m f_m(x))' \\ &= c_1 f_1'(x) + c_2 f_2'(x) + \dots + c_m f_m'(x) \end{aligned} \quad (2.17)$$

The proof of this rule is done by induction. For $m = 1$, it coincides with rule (2.16). Then, using rule (2.15), we have:

$$\begin{aligned} & (c_1 f_1(x) + \dots + c_m f_m(x))' \\ &= (c_1 f_1(x) + \dots + c_{m-1} f_{m-1}(x))' + (c_m f_m(x))' \\ &= c_1 f_1'(x) + \dots + c_{m-1} f_{m-1}'(x) + c_m f_m'(x). \end{aligned}$$

Here we have used the method of induction, that is, we assumed the rule is valid for $m - 1$ functions. Thus, rule (2.17) is proven.

Using rules (2.17), (2.13), and (2.14), we can now find the derivative of the polynomial (2.1). We have:

$$\begin{aligned} & (a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n)' \\ &= a_0 (x^n)' + a_1 (x^{n-1})' + \dots + a_{n-1} (x)' + (a_n)' \\ &= n a_0 x^{n-1} + (n-1) a_1 x^{n-2} + \dots + a_{n-1}. \end{aligned}$$

Thus, the rule for finding the derivative of the polyno-

mial (2.1) is finally written as:

$$\begin{aligned} & \left(a_0 x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n \right)' \\ &= a_0 x^n + n a_0 x^{n-1} + (n-1) a_1 x^{n-2} \\ &+ \dots + a_{n-1} \end{aligned} \tag{2.18}$$

3 Maximum and Minimum.

Rolles Theorem and Lagranges Formula

The very definition of the derivative suggests that it may serve as a useful tool for studying a function. For example, if it is known that the derivative $f'(x)$ of a function $f(x)$ is positive at a point x , then it is intuitively clear that the function $f(x)$ increases in the vicinity of this point. This is evident, in particular, from the geometric meaning of the derivative: *the tangent at the corresponding point on the graph of the function $f(x)$ is directed upwards.*

The situation is similar when the derivative is negative. In this case, it is intuitively clear that the function

$f(x)$ decreases near the point x . The *tangent to the graph of the function $f(x)$ at the corresponding point is directed downwards*. Let us now refine these properties of the derivative.

Positivity and Negativity of the Derivative

Let us recall formulas (1.31) and (1.30) from Section 1. We have:

$$f(\xi) - f(x) = k(\xi) (\xi - x) \quad \text{and} \quad (3.1)$$

$$\lim_{\xi \rightarrow x} k(\xi) = f'(x). \quad (3.2)$$

If $f'(x) \neq 0$, then for ξ sufficiently close to x , the number $k(\xi)$ has the same sign as $f'(x)$. More precisely, there exists a sufficiently small positive number ε such that

$$|\xi - x| < \varepsilon \quad (3.3)$$

implies that the sign of $k(\xi)$ coincides with the sign of $f'(x)$.

The sign of the right-hand side of equation (3.1) depends on the signs of its factors, i.e., the signs of the numbers $k(\xi)$ and $\xi - x$. To cover all four possible cases briefly, we choose two values ξ_1 and ξ_2 satisfying condition (3.3), with ξ_1 to the left of x , and ξ_2 to the

right of x . Thus, the numbers ξ_1 and ξ_2 satisfy:

$$x - \varepsilon < \xi_1 < x < \xi_2 < x + \varepsilon. \quad (3.4)$$

Recalling that the sign of $k(\xi)$ matches that of the derivative $f'(x)$, and taking into account the signs of the right-hand side of equation (3.1), we obtain the following two results:

$$\text{If } f'(x) > 0, \text{ then } f(\xi_1) < f(x) < f(\xi_2), \quad (3.5)$$

$$\text{If } f'(x) < 0, \text{ then } f(\xi_1) > f(x) > f(\xi_2). \quad (3.6)$$

In words, the results (3.5) and (3.6) can be described as follows:

§ When $f'(x) > 0$, the function to the left of the point x is less than at the point x , and to the right of the point x , it is greater than at the point x . In other words, the function is *increasing* at the point x .

§ When $f'(x) < 0$, the function to the left of the point x is greater than at the point x , and to the right of the point x , it is less than at the point x . In other words, the function is *decreasing* at the point x .

Note that if the function is increasing at the point x , i.e., if the inequalities (3.5) are satisfied, then the difference quotient $(f(\xi) - f(x))/(\xi - x)$ is positive ,

but as $\xi \rightarrow x$, while remaining positive, it may tend to zero. Thus, the derivative at a point of increase is not necessarily positive; it is only non-negative:

$$f'(x) \geq 0. \quad (3.7)$$

Similarly, at a point of decrease (see (6)), the derivative is not necessarily negative; it may also be zero. Therefore, at a point of decrease, the following inequality holds:

$$f'(x) \leq 0. \quad (3.8)$$

Maximum and Minimum

It is said that a *function has a local maximum at a point x* if, in all points sufficiently close to x , its values do not exceed the value of the function at point x . More precisely, there exists a sufficiently small positive number ϵ such that

$$|\xi - x| < \epsilon \text{ then we have } f(\xi) \leq f(x). \quad (3.9)$$

Similarly, it is said that a *function has a local minimum at a point x* if, for all values of the argument close to x , the function has values not less than at the point x . More precisely, there exists a positive number ϵ such that

$$|\xi - x| < \epsilon \text{ then we have } f(\xi) \geq f(x). \quad (3.10)$$

Usually, the word “local” is omitted, and one simply speaks of the maxima and minima of the function.

It turns out that *at the points of maximum and minimum, the derivative of the function $f(x)$ equals zero*:

$$f'(x) = 0. \quad (3.11)$$

Indeed, at the point of maximum, the function $f(x)$ cannot have a positive derivative, because in the case of a positive derivative to the right of x , the function is greater than at the point x (see (3.5)), and, therefore, the point x cannot be a maximum. Similarly, in the case of a maximum, the function $f(x)$ cannot have a negative derivative, because in this case, to the left of point x , the function has values greater than at point x (see (3.6)). The only remaining possibility is $f'(x) = 0$, i.e., the equality (3.11) holds.

Similarly, at the point of minimum, the function $f(x)$ cannot have a positive derivative, because then, to the left of point x , its values are smaller than at the point x (see (3.5)). Likewise, it cannot have a negative derivative, because then, to the right of point x , it has a smaller value than at the point x (see (3.6)). Therefore, the only remaining possibility is $f'(x) = 0$, i.e., the equality (3.11) holds.

Thus, to find the values of the argument for which the function has a maximum or minimum, one should

consider all values of x for which the equality (3.11) holds, and then further investigate the matter in more detail.

Rolle's Theorem

Rolle's Theorem

If the function $f(x)$ takes equal values at two different points of its domain x_1 and x_2 , that is,

$$f(x_1) = f(x_2), \quad (3.12)$$

and if the function $f(x)$ is defined on the entire interval $[x_1, x_2]$, then there exists a point θ inside this interval such that

$$f'(\theta) = 0. \quad (3.13)$$

The term “inside the interval” means that θ does not coincide with either endpoint x_1 or x_2 , but lies strictly *between* them.

Let us prove this assertion. If the function $f(x)$ is constant on the entire interval $[x_1, x_2]$, then in any interior point of the interval, the following relation holds: $f'(\theta) = 0$ (see Section 2, (2.14). If the function

$f(x)$ is not constant on $[x_1, x_2]$, then at least one of the following two cases occurs:

- Case 1.** There exist some points in the interval $[x_1, x_2]$ where the function $f(x)$ is *greater* than at the endpoints.
- Case 2.** There exist some points in the interval $[x_1, x_2]$ where the function $f(x)$ is *smaller* than at the endpoints.

In the first case, the function $f(x)$ has a maximum at some interior point θ of the interval $[x_1, x_2]$, and then at this point the equality (3.13) holds (see (3.11)). In the second case, the function $f(x)$ attains its minimum at an interior point θ of the interval $[x_1, x_2]$, and then again the equality (3.13) holds (see (3.11)). Thus, Rolle's Theorem is proved.

The following Lagrange formula is a direct consequence of Rolle's Theorem.

Lagrange's Formula of Finite Increments of a Function

Lagrange's Formula

If x_1 and x_2 are two distinct values of the argument of the function $f(x)$, and the function $f(x)$ is defined on the entire interval $[x_1, x_2]$, then there exists a value θ , lying inside the interval $[x_1, x_2]$, such that the equality

$$f(x_2) - f(x_1) = f'(\theta) (x_2 - x_1) \quad (3.14)$$

holds.

To prove this statement, let us construct the linear function

$$g(x) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \cdot x, \quad (3.15)$$

and show that the function

$$f(x) - g(x) \quad (3.16)$$

has equal values at the points x_1 and x_2 , i.e., satisfies

the conditions of Rolles Theorem. Indeed, we have

$$\begin{aligned} g(x_2) - g(x_1) &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \cdot x_2 - \frac{f(x_2) - f(x_1)}{x_2 - x_1} \cdot x_1 \\ &= f(x_2) - f(x_1). \end{aligned} \quad (3.17)$$

Next, we have

$$\begin{aligned} (f(x_2) - g(x_2)) - (f(x_1) - g(x_1)) \\ &= (f(x_2) - f(x_1)) - (g(x_2) - g(x_1)) \quad (3.18) \\ &= 0. \end{aligned}$$

(see (3.17)). Thus, we obtain

$$f(x_2) - g(x_2) = f(x_1) - g(x_1), \quad (3.19)$$

i.e., the function (3.16) has equal values at the ends of the interval $[x_1, x_2]$, and therefore, by Rolles Theorem, there exists a value θ inside the interval (x_1, x_2) such that

$$(f(x) - g(x))' = 0 \quad \text{at} \quad x = \theta. \quad (3.20)$$

Next, we have

$$g'(x) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}. \quad (3.21)$$

From equations (3.20) and (3.21), we get

$$0 = f'(\theta) - g'(\theta) = f'(\theta) - \frac{f(x_2) - f(x_1)}{x_2 - x_1}. \quad (3.22)$$

Multiplying the last relation by $x_2 - x_1$, we obtain the required relation (3.14). Thus, the Lagrange formula is proven.

The Lagrange formula is a powerful tool for studying a function, or, equivalently, its graph. Let us derive two important consequences from it.

If on the interval $[x_1, x_2]$, where $x_1 < x_2$, the function $f(x)$ has a positive derivative at all points, except possibly at the endpoints, then the function $f(x)$ is increasing on the entire interval $[x_1, x_2]$. More precisely, if a_1 and a_2 are two values of the argument lying within the interval $[x_1, x_2]$, and $a_1 < a_2$, then we have

$$f(a_1) < f(a_2). \quad (3.23)$$

Indeed, by virtue of the Lagrange formula (3.14), we have

$$f(a_2) - f(a_1) = f'(\theta) \cdot (a_2 - a_1), \quad (3.24)$$

where θ is an interior point of the interval $[a_1, a_2]$, and hence also an interior point of the interval $[x_1, x_2]$. Thus, the right-hand side of equation (3.23) is positive, proving our claim (3.22).

If on the interval $[x_1, x_2]$, where $x_1 < x_2$, the function $f(x)$ has a negative derivative everywhere except possibly at the ends of the interval $[x_1, x_2]$, then on the entire interval $[x_1, x_2]$ it is decreasing. More precisely,

if a_1 and a_2 are two values of the argument located in the interval $[x_1, x_2]$, and $a_1 < a_2$, then

$$f(a_2) < f(a_1). \quad (3.25)$$

Indeed, by virtue of formula (3.14) we have

$$f(a_2) - f(a_1) = f'(\theta)(a_2 - a_1), \quad (3.26)$$

where θ is an interior point of the interval $[a_1, a_2]$, i.e., an interior point also of the interval $[x_1, x_2]$. Thus, the right-hand side of expression (3.25) is negative, and therefore, the assertion (3.24) is proven.

The Second Derivative

The derivative $f'(x)$ of the function $f(x)$ is itself a function, and therefore we can take its derivative, $(f'(x))'$. This is called the *second derivative* of the function $f(x)$ and is denoted by $f''(x)$. Thus,

$$f''(x) = (f'(x))' \quad (3.27)$$

The functions $f'(x)$ and $f''(x)$ are called the first and second derivatives of the function $f(x)$. Analogously, one can define derivatives of any order of the function $f(x)$, but we shall make use only of the first and second derivatives.

Distinguishing Between a Maximum and a Minimum

For a function $f(x)$ to have a maximum or a minimum at $x = x_0$, it is necessary that

$$f'(x_0) = 0, \quad (3.28)$$

(see (3.11)). But this equation is *not a sufficient condition* for the function $f(x)$ to have a maximum or minimum at the point $x = x_0$, and furthermore, it does not allow us to distinguish between a maximum and a minimum. It turns out that a *sufficient condition* is given by the second derivative, namely, if

$$f''(x_0) \neq 0, \quad (3.29)$$

then, provided condition (3.27) holds, the function $f(x)$ has either a maximum or a minimum at the point $x = x_0$. The sign of the quantity $f''(x_0)$ allows us to distinguish between a maximum and a minimum, namely:

$$f''(x_0) < 0 \text{ then we have a maximum} \quad (3.30)$$

$$f''(x_0) > 0 \text{ then we have a minimum} \quad (3.31)$$

Let us prove this statement. If inequality (3.29) holds, then the first derivative $(f'(x))'$ of the function $f'(x)$ is

negative. Moreover, the function $f'(x)$ is equal to zero at $x = x_0$ (see (3.27)). Thus,

$$\text{For } x < x_0, \text{ we have } f'(x) > 0, \quad (3.32)$$

$$\text{For } x > x_0, \text{ we have } f'(x) < 0. \quad (3.33)$$

Hence, as x approaches x_0 from the left, the function $f(x)$ increases, and as x moves away from x_0 to the right, it decreases. Therefore, we have a *maximum*.

Similarly, if inequality (3.30) holds, then the first derivative $(f'(x))'$ of the function $f'(x)$ is positive, and $f'(x)$ is equal to zero at $x = x_0$. Thus,

$$\text{For } x < x_0, \text{ we have } f'(x) < 0, \quad (3.34)$$

$$\text{For } x > x_0, \text{ we have } f'(x) > 0. \quad (3.35)$$

Hence, as x approaches x_0 from the left, the function $f(x)$ decreases, and as x moves away from x_0 to the right, it increases. Therefore, we have a *minimum*.

4 Study Of Functions

Let us recall the definition of the second derivative $f''(x)$ of the function $f(x)$. We have

$$f''(x) = (f'(x))' \quad (4.1)$$

(see Section 3, (3.26)). The functions $f'(x)$ and $f''(x)$ are called the first and second derivatives of the function $f(x)$, respectively. Let us now apply the results from Section 3 to study certain functions defined by polynomials. First, consider the function

$$y = f(x) = x^3 - p x, \quad (4.2)$$

where p is a constant. The graph of this function is called a *cubic parabola*.

Let us first note some very special but quite striking properties of the cubic parabola. The cubic parabola is

centrally symmetric with respect to the origin. Indeed, if the point (x, y) lies on the cubic parabola — that is, the values x and y satisfy equation (4.2) — then the point $(-x, -y)$ also satisfies this equation. Namely, we have

$$(-y) = (-x)^3 - p(-x). \quad (4.3)$$

Thus, along with the point (x, y) , the point symmetric to it with respect to the origin, $(-x, -y)$, also lies on the curve $f(x)$.

Let us now find the points where the curve f intersects the x -axis, that is, the roots of the equation

$$x^3 - p x = 0. \quad (4.4)$$

This equation has three roots:

$$x = 0, \quad x = \pm\sqrt{p}. \quad (4.5)$$

The latter two roots are imaginary when $p < 0$, and thus have no geometric meaning. When $p > 0$, all three roots in (4.5) are distinct, and therefore the curve f intersects the x -axis at three points. When $p = 0$, all three roots merge into a single triple root at $x = 0$.

The derivative of the function given in equation (4.2) is

$$f'(x) = 3x^2 - p. \quad (4.6)$$

By studying the sign of this function for various values of x , we can divide the curve f into intervals where the function $f(x)$ is increasing or decreasing, and identify the points of maximum and minimum. To do both, we must find the roots of the equation

$$f'(x) = 3x^2 - p = 0 \quad (4.7)$$

When $p < 0$, the derivative $f'(x)$ (see equation (4.6)) is positive for all values of x . Therefore, the function $f(x)$ is increasing throughout the entire domain: $-\infty < x < +\infty$.

When $p = 0$, the derivative $f'(x)$ is also positive for all $x \neq 0$ (see again equation (4.6)). Thus, the function is increasing on both intervals: $-\infty < x < 0$ and $0 < x < +\infty$.

Moreover, since the function x^3 is negative on the first of these intervals and positive on the second, it follows that the function $f(x)$ increases over the entire real line: $-\infty < x < +\infty$.

At the point $x = 0$, where $f'(x) = 0$, the function $f(x)$ still increases. Thus, at $x = 0$, although the derivative vanishes, the function x^3 does not attain either a maximum or a minimum. This shows that the condition (3.11) of Section

refch-03, which is necessary for a function $f(x)$ to have a maximum or minimum at a point x , is not sufficient.

In the case of a positive p , equation (4.7) has two roots:

$$x_1 = -\sqrt{p/3}, \quad x_2 = \sqrt{p/3} \quad (4.8)$$

We must check whether the function $f(x)$ has local maxima or minima at the points x_1 and x_2 . These points divide the domain of x into the following intervals:

$$-\infty < x \leq x_1, \quad x_1 \leq x \leq x_2, \quad x_2 < x < +\infty \quad (4.9)$$

On the first of these intervals, the function $f'(x)$ is positive; on the second, it is negative; and on the third, it is again positive. Thus, the function $f(x)$ increases on the first interval, decreases on the second, and increases on the third. From this, it follows that the point x_1 is a point of local maximum, and the point x_2 is a point of local minimum.

Hence, the cubic parabola takes on three essentially different forms depending on the value of p :

§ First case, when $p < 0$,

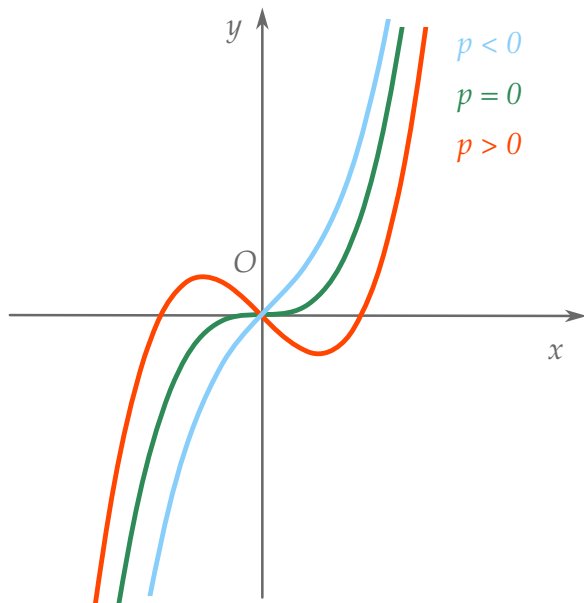
§ second case, when $p = 0$,

§ third case, $p > 0$.

In Figure 2, the cubic parabola is shown for all three cases. Let us now consider the equation

$$x^3 - px = c. \quad (4.10)$$

4 Study Of Functions



Geometrically, it is clear that when $p < 0$, this equation has only one real root. When $p = 0$, it also has only one real root, except in the special case $c = 0$, when it has a triple root at $x = 0$. When $p > 0$, the equation (4.10) has three real roots if

$$f(x_2) \leq c \leq f(x_1), \quad (4.11)$$

Figure 2: The cubic parabola $x^3 - px$ for different values of p .

in the extreme cases on the boundary of interval (4.11), there is one simple root and one double root. Outside the interval (4.11), the equation (4.10) has only one real root.

The cubic parabola defined by equation (4.2) always passes through the origin. The tangent of the angle of inclination at the origin is given by

$$f'(0) = -p \quad (4.12)$$

Thus, the equation of the tangent line at the origin is

$$y = g(x) = -px \quad (4.13)$$

This tangent line divides the entire plane into two parts: the upper part, lying above the line, and the lower part, lying below it. An arbitrary point (x^*, y^*) lies above the line (4.13) if

$$y^* > -px^* \quad (4.14)$$

and lies below the line if

$$y^* < -px^* \quad (4.15)$$

Let us now determine in which of these two half-planes the point (x, y) belonging to the cubic parabola (4.2) lies, that is, the point satisfying that equation. To answer this, we need to compare the value

$$x^3 - px \quad (4.16)$$

with the value

$$-px \quad (4.17)$$

That is, we compare the actual ordinate of the point on the curve with the ordinate of the tangent line at the same abscissa.

It is clear that for $x < 0$, the value given by equation (4.16) is less than the value given by equation (4.17), and for $x > 0$, the value given by equation (4.16) is greater than the value given by equation (4.17).

Thus, for $x < 0$, the point (x, y) on the cubic parabola satisfies condition (4.15), meaning it lies below the tangent line, while for $x > 0$, the point (x, y) satisfies condition (4.14), meaning it lies above the tangent line.

Therefore, the cubic parabola passes from one side of the tangent line to the other side at the origin.

This phenomenon of a curve passing from one side of the tangent line to the other near the point of tangency is of general interest. Let us now investigate this phenomenon.

Inflection Point

Let L be the graph of some function $f(x)$, and a be a certain point on the line L with the abscissa x_0 , and K

be the tangent to L at point a (see Figure 3). If the line L passes from one side of the tangent line to the other at point a , then point a is called the *inflection point* of the curve L .

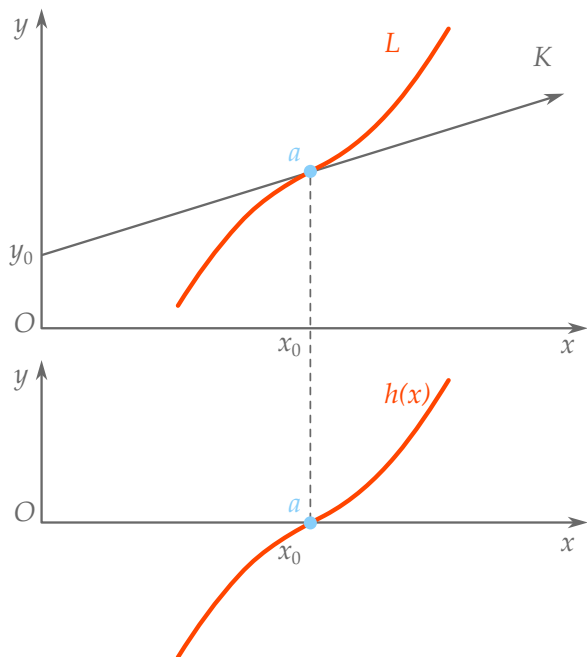


Figure 3: Geometric meaning of the inflection point of a function.

It turns out that for an inflection point, the following

condition holds:

$$f''(x_0) = 0. \quad (4.18)$$

Let us prove this statement. The equation of the tangent line K can be written as:

$$y = g(x) = f'(x_0) \cdot x + y_0. \quad (4.19)$$

Here, $f'(x_0)$ is the tangent of the angle of the tangent line to the x-axis, and y_0 is some constant determined by the condition:

$$f(x_0) - g(x_0) = 0. \quad (4.20)$$

This condition expresses the fact that the tangent at point a to the curve L passes through point a . Let us consider the function:

$$h(x) = f(x) - g(x). \quad (4.21)$$

This function satisfies two conditions:

$$h(x_0) = 0, \quad h'(x_0) = 0. \quad (4.22)$$

Thus, the graph of the function

$$y = h(x) \quad (4.23)$$

is tangent to the x-axis at $x = x_0$ (see Figure 3). If a is an inflection point, the graph of the function $h(x)$

at x_0 crosses from one side of the x -axis to the other. We will now prove that $h''(x_0) = 0$. We will proceed by contradiction. Suppose $h''(x_0) > 0$. Since $h'(x_0) = 0$, the function $h(x)$ has a local minimum at x_0 (see Section 3 equation (3.30)), and thus for all x sufficiently close to x_0 , we have:

$$h(x) \geq h(x_0) = 0.$$

Therefore, the graph of the function $h(x)$ does not cross the x -axis at the point x_0 . The case $h''(x_0) \leq 0$ is treated similarly. In that case, the function $h(x)$ has a local maximum at x_0 , and hence: $h(x) \leq h(x_0) = 0$. Therefore, even when $h''(x_0) < 0$, the graph of $h(x)$ does not cross the x -axis at x_0 . Thus, the only possibility that remains is $h''(x_0) = 0$.

Next, we have

$$0 = h''(x_0) = f''(x_0) - g''(x_0) = f''(x_0), \quad (4.24)$$

since $g''(x) = 0$. Thus, formula (4.18) proves the earlier claim that the second derivative at the inflection point is zero.

However, this condition is *not sufficient* for the point x_0 to be an inflection point of the graph of the function $f(x)$.

Let us now return to the cubic parabola. We compute the second derivative of the function:

$$f''(x) = (x^3 - px)'' = 6x, \quad (4.25)$$

We have already established that the origin is an inflection point of the cubic parabola (4.2). The expression ((4.25)) for the second derivative of the function (4.2) shows that the origin is the only inflection point of the cubic parabola, since the second derivative ((4.25)) vanishes only at $x = 0$.

Exercises

Below are three problems for independent work. The first one is easy, while the second and third are quite serious mathematical problems that require significant effort to solve independently.

Problem 1. Change the length scales on the coordinate axes of the system in which equation (4.2) is given, i.e., introduce new coordinates x_1 and y_1 instead of the old coordinates x and y , by setting

$$x = kx_1, \quad y = ly_1, \quad (4.26)$$

where k and l are positive real numbers. Do this in such a way that after the coordinate transformation,

equation (4.2) takes one of the following three forms:

$$y_1 = x_1^3 - x_1, \quad y_1 = x_1^3, \quad y_1 = x_1^3 + x_1. \quad (4.27)$$

Problem 2. Study the function

$$y = f(x) = x^3 + a_1x^2 + a_2x + a_3. \quad (4.28)$$

To do this, compute the derivative $f'(x)$, and with its help divide the entire interval of variation of x , that is $-\infty < x < +\infty$, into parts where the function increases and decreases. Then, instead of the old coordinates x and y , introduce new coordinates x_1 and y_1 , related to the old ones by the relations

$$x = x_1 + \alpha, \quad y = y_1 + \beta. \quad (4.29)$$

Such a coordinate transformation represents a parallel shift of the original coordinate system. This shift should be done so that in the new coordinate system, the equation of the curve takes the form (4.2). This can be achieved in two ways: either by directly selecting the values of α and β such that the resulting equation takes the form (4.2), or by finding the inflection point of the graph of the function L (4.28) and shifting the origin of the coordinate system to this inflection point. Determine the formula

$$p = p(a_1, a_2, a_3), \quad (4.30)$$

expressing the coefficient p in the new coordinate system in terms of the original coefficients. Since the equation

$$f(x) = x^3 + a_1x^2 + a_2x + a_3 = 0, \quad (4.31)$$

may have only one real root when $p(a_1, a_2, a_3) < 0$ (see (4.30)), may have only one real root or one triple real root when $p = 0$, and may have three real roots when $p > 0$, determine under what conditions, in the last case, three real roots occur.

Problem 3. Study the function

$$f(x) = x^4 + b_1x^3 + b_2x^2 + b_3x + b_4, \quad (4.32)$$

and its graph. To do this, compute the derivative $f'(x)$ of the function (4.32), and using the results obtained in Problem 2, qualitatively analyse the possible behaviour of the graph of the function (4.32), particularly the number of possible maxima and minima, and determine how this number depends on the coefficients b_1, b_2, b_3, b_4 of the function (4.32). Also, find the inflection points and express their coordinates in terms of the coefficients b_1, b_2, b_3, b_4 .

5 Derivatives of Trigonometric Functions and Some Differentiation Rules

Here we shall first compute the derivatives of the trigonometric functions $\sin x$ and $\cos x$, where the angle x is measured in radians, not in degrees. In this computation, we shall use a certain fact, which is stated without proof. The proof is cumbersome and uninteresting, and it is easy to believe the fact. The fact is as follows.

Let K be a circle, and let a and b be two points on it, not located diametrically opposite each other. We denote the length of the smaller arc ab of the circle K by $s(ab)$, and the length of the chord ab by $l(ab)$. It is

evident that

$$s(ab) > l(ab). \quad (5.1)$$

We accept without proof that the ratio $l(ab)/s(ab)$ tends to unity as the points a and b approach one another indefinitely, that is, as $s(ab) \rightarrow 0$. In formulaic terms, this can be written as:

$$\lim_{s(ab) \rightarrow 0} \frac{l(ab)}{s(ab)} = 1. \quad (5.2)$$

From this unproven assertion, we shall now rigorously derive another result which we shall use. To this end, let us choose in the coordinate plane a circle K of radius 1 with its centre at the origin O (see Figure 4).

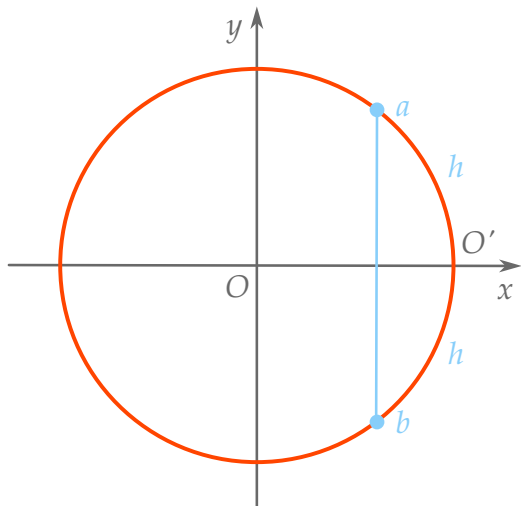


Figure 4: A unit circle K .

Let us denote by O' the rightmost point of the circle, i.e., the point of intersection of the circle with the positive half of the x -axis. From the point O' , lay off upwards along the circle an arc of length $h < \pi/2$, and denote its endpoint by b . Similarly, from the point O' , lay off downwards an arc of length h , and denote its endpoint by a . Then we have:

$$l(ab) = 2 \sin h, \quad s(ab) = 2h. \quad (5.3)$$

5 Derivatives of Trigonometric Functions

Thus, from equation (5.2) we obtain:

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = \lim_{2h \rightarrow 0} \frac{2 \sin h}{2h} = 1.$$

Thus, we finally obtain the desired relation:

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1. \quad (5.4)$$

This formula is proven for positive h , but it also holds for negative h , since changing the sign of h changes the sign of $\sin h$ as well.

In calculating the derivatives of $\sin x$ and $\cos x$, we shall make use of two well-known trigonometric identities:

$$\sin \alpha - \sin \beta = 2 \sin \left(\frac{\alpha - \beta}{2} \right) \cos \left(\frac{\alpha + \beta}{2} \right) \quad (5.5)$$

$$\cos \alpha - \cos \beta = -2 \sin \left(\frac{\alpha - \beta}{2} \right) \sin \left(\frac{\alpha + \beta}{2} \right) \quad (5.6)$$

Let us find the preliminary limit (see Section 1 (1.21)) for the function $\sin x$. We have:

$$\frac{\sin \xi - \sin x}{\xi - x} = \frac{\sin \left(\frac{\xi - x}{2} \right)}{\frac{\xi - x}{2}} \cos \left(\frac{\xi + x}{2} \right).$$

Taking the limit of this expression as $\xi \rightarrow x$, and using equation (5.4), we obtain:

$$\sin' x = \cos x \quad (5.7)$$

For the function $\cos x$, we have:

$$\frac{\cos \xi - \cos x}{\xi - x} = \frac{\sin\left(\frac{\xi - x}{2}\right)}{\frac{\xi - x}{2}} \sin\left(\frac{\xi + x}{2}\right).$$

Taking the limit of this expression as $\xi \rightarrow x$, and using equation (5.4), we obtain:

$$\cos' x = -\sin x \quad (5.8)$$

Derivative of a Product and a Quotient

We already know how to find the derivative of the sum of two functions (see Section 2, equation (2.15)). It is obviously very important to be able to find the derivative of the *product* of two functions and the *quotient* (ratio) of two functions.

Let us denote two functions of the variable x by $u(x)$ and $v(x)$, and we shall look for the derivatives of the functions $u(x) \cdot v(x)$ and $u(x)/v(x)$.

In finding these derivatives, we shall use the definition of the derivative (see Section 1, equation (1.21)). Thus, to find the derivative of the product, we first compute the expression:

$$\begin{aligned} u(\xi) v(\xi) - u(x) v(x) &= u(\xi) v(\xi) - u(x) v(\xi) + u(x) v(\xi) - u(x) v(x) \\ &= (u(\xi) - u(x)) v(\xi) + u(x) (v(\xi) - v(x)). \end{aligned}$$

Therefore, we obtain:

$$\begin{aligned} (u(x) v(x))' &= \lim_{\xi \rightarrow x} \frac{u(\xi) v(\xi) - u(x) v(x)}{\xi - x} \\ &= \lim_{\xi \rightarrow x} \left[\frac{(u(\xi) - u(x)) v(\xi)}{\xi - x} \right] + \lim_{\xi \rightarrow x} \left[\frac{u(x) (v(\xi) - v(x))}{\xi - x} \right] \\ &= u'(x) v(x) + u(x) v'(x). \end{aligned}$$

Hence, we finally obtain the important formula:

$$(u(x) v(x))' = u'(x) v(x) + u(x) v'(x). \quad (5.9)$$

We proceed in exactly the same way to find the derivative of a *quotient* $u(x)/v(x)$. For this, we must first compute the expression:

$$\begin{aligned} \left(\frac{u(\xi)}{v(\xi)} \right) - \left(\frac{u(x)}{v(x)} \right) &= \frac{u(\xi) v(x) - v(\xi) u(x)}{v(\xi) v(x)} \\ &= \frac{u(\xi) v(x) - u(x) v(x) + u(x) v(x) - v(\xi) u(x)}{v(\xi) v(x)} \\ &= \frac{(u(\xi) - u(x)) v(x) - (v(\xi) - v(x)) u(x)}{v(\xi) v(x)}. \end{aligned}$$

Then, taking the limit as $\xi \rightarrow x$, we will be able to derive the rule for the derivative of a quotient.

$$\begin{aligned} \left(\frac{u(x)}{v(x)} \right)' &= \lim_{\xi \rightarrow x} \frac{\frac{u(\xi) - u(x)}{v(\xi) - v(x)}}{\xi - x} \\ &= \lim_{\xi \rightarrow x} \frac{\frac{u(\xi) - u(x)}{\xi - x} v(x)}{v(\xi)v(x)} - \lim_{\xi \rightarrow x} \frac{\frac{v(\xi) - v(x)}{\xi - x} u(x)}{v(\xi)v(x)} \\ &= \frac{u'(x) v(x) - v'(x) u(x)}{(v(x))^2} \end{aligned}$$

Thus, we finally obtain the fundamental formula for the derivative of a quotient of two functions:

$$\left(\frac{u(x)}{v(x)} \right)' = \frac{u'(x) v(x) - v'(x) u(x)}{(v(x))^2} \quad (5.10)$$

The Derivative of $\tan x$ function

To find the derivative of the function $\tan x$, we use the quotient rule (5.10). Namely, let

$$\tan x = \frac{\sin x}{\cos x}$$

Then, according to rule (5.10) and taking into account (5.7) and (5.8), we have:

$$\begin{aligned}\tan' x &= \frac{\sin' x \cos x - \cos' x \sin x}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}.\end{aligned}$$

Thus, we finally obtain:

$$\tan' x = \frac{1}{\cos^2 x} = 1 + \tan^2 x. \quad (5.11)$$

Derivative of a Composite Function

We will proceed from two given functions, $\varphi(x)$ of the variable x , $\psi(y)$ of the variable y . Let

$$y = \varphi(x).$$

Substituting this expression for y into the function $\psi(y)$, we obtain the function

$$f(x) = \psi(y) = \psi(\varphi(x)). \quad (5.12)$$

This new function $f(x)$, formed from two functions $\psi(y)$ and $\varphi(x)$, is called a *composite function*. Our task is to find the derivative $f'(x)$ of the composite function

5 Derivatives of Trigonometric Functions

$f(x)$. We will use the definition of the derivative (see Section 1, formula (1.21)). Let

$$\eta = \varphi(\xi). \quad (5.13)$$

Then, as

$$\xi \rightarrow x, \text{ we have } \eta \rightarrow y. \quad (5.14)$$

According to equation (1.31) from Section 1, we have:

$$\begin{aligned} \psi(\eta) - \psi(y) &= k(\eta) (\eta - y), \text{ where} \\ \lim_{\eta \rightarrow y} k(\eta) &= \psi'(y). \end{aligned}$$

Now we have,

$$\begin{aligned} \frac{f(\xi) - f(x)}{\xi - x} &= \frac{k(\eta) (\eta - y)}{\xi - x} \\ &= \frac{k(\eta) (\varphi(\xi) - \varphi(x))}{\xi - x} \end{aligned}$$

Passing in this ratio to the limit at $\xi \rightarrow x$, we obtain

$$f'(x) = \psi'(y) \varphi'(x) = \psi'(\varphi(x)) \varphi'(x). \quad (5.15)$$

Thus, we finally get

$$(\psi(\varphi(x)))' = \psi'(\varphi(x)) \varphi'(x). \quad (5.16)$$

Let us explain the written formula in words: in order to obtain the derivative of the composite function $f(x) =$

$\psi(\varphi(x))$ we must first calculate the derivative of the function $\psi(y)$ with respect to the variable y , that is, find the function $\psi'(y)$, and then substitute $\psi(x)$ in place of y in the resulting expression. After this, we must multiply the resulting expression by $\varphi'(x)$.

Inverse Function

In mathematics and its applications, functions are often defined as solutions to equations. The most important case is the following.

Let us begin with a given function $\psi(y)$ of the variable y . The equation

$$\psi(y) = x, \quad (5.17)$$

connects the variables x and y , and it may happen that this equation is solvable for y . Since x appears in the equation, the solution with respect to y depends on x and is thus a function of x . We denote it by $\varphi(x)$. Thus, we have the identity

$$\psi(\varphi(x)) = x. \quad (5.18)$$

The function $\varphi(x)$ that satisfies this identity is called the *inverse of the function* $\psi(y)$.

Let us prove that the function $\psi(y)$ is the inverse of the function $\varphi(x)$. To do so, substitute into the function

$\varphi(x)$ the left- and right-hand sides of equality (5.18), taking into account that $\varphi(x) = y$. Then we obtain

$$\varphi(\psi(y)) = y. \quad (5.19)$$

By the definition of the inverse function, formula (5.19) shows that the function $\varphi(y)$ is the inverse of the function $\varphi(x)$. Thus, the functions $\psi(y)$ and $\varphi(x)$ are *mutually inverse*.

The equation $\psi(y) = x$ may be unsolvable with respect to y , for example, when $\psi(y)$ is a constant. In this case, it does not define y as a function of x . Equation (5.17) may be solvable but not uniquely. For example, the equation $y^2 = x$ has two solutions $y = +\sqrt{x}$ and $y = -\sqrt{x}$.

Let us determine under what conditions equation (5.17) is solvable and defines a function $y = \varphi(x)$. Assume that the derivative of the function $\psi(y)$ maintains its sign on the interval $b_1 \leq y \leq b_2$, vanishing possibly only at its endpoints. Let

$$a_1 = \psi(b_1); \quad a_2 = \psi(b_2). \quad (5.20)$$

If the derivative of the function $\psi(y)$ on the interval $b_1 < y < b_2$ is positive, then the function $\psi(y)$ increases on this interval, so that $a_1 < a_2$. Otherwise, the function $\psi(y)$ decreases and $a_1 > a_2$. It turns out that there

is a function $\varphi(x)$ defined in the section (a_1, a_2) and, moreover, only one inverse to the function $\psi(y)$, so that identity (5.18) is fulfilled. In this case, the derivative of the function $\varphi(x)$ exists, with the possible exception of only the edge points a_1, a_2 , in which it can go to infinity. This derivative is determined by the formula

$$\varphi'(x) = \frac{1}{\psi'(y)} = \frac{1}{\psi'(\varphi(x))}. \quad (5.21)$$

We will prove this by constructing the graph L of the function $y = \psi(x)$ in an unconventional manner, by plotting the value of its argument y along the ordinate axis and the value of the function $x = \psi(y)$ along the abscissa axis. For a geometric solution to the equation $\psi(y) = x$ with respect to y , we must, for a given value of x , draw a line parallel to the ordinate axis through the point x on the abscissa axis, selecting x from the interval (a_1, a_2) . The drawn line will intersect the curve L $y = \varphi(x)$ at exactly one point because the function $\psi(y)$ is either increasing or decreasing. The ordinate $y = \varphi(x)$ of this intersection point is the solution to the equation $\psi(y) = x$ with respect to y .

Thus, the line L is the graph of the function $y = \varphi(x)$ in the usual sense. Since the function $\varphi(y)$ is differentiable, the line L has a tangent at each of its points $a = (x, y)$. Let us denote this tangent by K . Thus, the graph of the function $y = \varphi(x)$ has a tangent

at each of its points A . Therefore, the function $\varphi(x)$ has a derivative for every value of x , except in the case where the tangent line K at point $a = (x, y)$ is vertical. In that case, the derivative $\varphi'(x)$ tends to infinity, but this can only happen at the ends of the segment $[a_1, a_2]$, if $\varphi'(b_1)$ or $\varphi'(b_2)$ are equal to zero.

Now that we know the function $\varphi(x)$ has a derivative, we can find it by differentiating the identity (5.18) with respect to x , treating the left-hand side as a composite function. Thus, we get

$$\psi'(y) \varphi'(x) = 1,$$

from which the formula (5.21) follows. Since the tangent line K at point a of the line L is vertical if and only if it is parallel to the y -axis, i.e., when $\psi'(y) = 0$, the derivative $\varphi'(x)$ tends to infinity only when

$$\psi'(\varphi(x)) = 0,$$

which is expressed by the formula (5.21).

Remark. When constructing a function inverse to the function $\psi(y)$, it sometimes happens that either one or both of the boundary values b_1 and b_2 of the considered interval are not admissible values of the argument of the function $\psi(y)$. In that case, they should not be considered. In particular, it may happen that one or both of the boundary values are infinite.

If one or both boundary values are not admissible arguments of the function $\psi(y)$, then the numbers a_1 and a_2 must not be defined by formulas (5.20), but should instead be considered as boundary values approached by the function $\psi(y)$. In this case, one or both of the quantities a_1 and a_2 may turn out to be infinite. In all such cases, the inverse function exists, and formula (5.21) remains valid.

Derivative of a Rational Power of x .

Using the results obtained here, we compute the derivative of the function

$$y = f(x) = x^r, \quad (5.22)$$

where r is a rational number. Namely, we shall prove that

$$(x^r)' = rx^{r-1}. \quad (5.23)$$

Note that the function x^r is not defined at $x = 0$ when r is negative. Also, when $0 < r < 1$, its derivative becomes infinite at $x = 0$.

We prove formula (5.23). First consider the case when r is a negative integer. Then let $n = -r$ so that n is a natural number. We have

$$x^r = \frac{1}{x^n}.$$

Differentiating the right-hand side as a quotient, we obtain

$$\begin{aligned}(x^r)' &= \left(\frac{1}{x^n}\right)' = \frac{1' \cdot x^n - n \cdot x^{n-1}}{x^{2n}} \\ &= -nx^{n-1} = rx^{r-1}.\end{aligned}$$

Thus, for negative integer r , formula (5.23) is proved.

Let $r = 1/n$, where n is a natural number greater than one. The function $x^{1/n}$ is defined by the formula

$$y = x^{1/n} = +\sqrt[n]{x}. \quad (5.24)$$

Note that when n is even, the right-hand side is defined only for $x \geq 0$, and in that case $y \geq 0$. When n is odd, the function $x^{1/n}$ is defined for all values of x .

The function (5.24) is defined as the solution of the equation

$$\psi(y) = y^n = x, \quad (5.25)$$

i.e. as an inverse function. We shall compute its derivative using formula (5.21). When n is even, we assume $y \geq 0$, so that y varies over the interval $0 \leq y < \infty$, and x varies over $0 < x < \infty$. The derivative $(y^n)' = ny^{n-1}$ on this interval of y is positive, except possibly at the endpoint $y = 0$. Therefore, the inverse function $y = \sqrt[n]{x}$ exists and has a derivative, which becomes infinite only at $x = 0$.

5 Derivatives of Trigonometric Functions

When n is odd, we allow y to be any real number. To prove the existence of the inverse function, we divide the entire interval $-\infty < y < \infty$ into two parts: $-\infty < y \leq 0$ and $0 \leq y < \infty$. On each of these intervals, the derivative $(y^n)' = ny^{n-1}$ is positive, except at the endpoint $y = 0$, where it becomes zero.

Therefore, on each of the two intervals, an inverse function exists. When considered over the union of these intervals, it is defined for all values of x . Its derivative also exists for all x except at $x = 0$, where the derivative of the inverse function becomes infinite. The derivative of the function $\phi(x) = x^{1/n}$ is computed using the formula

$$\begin{aligned}(x^{1/n})' &= \frac{1}{ny^{n-1}} = \frac{1}{\frac{n-1}{n}x} \\&= \frac{1}{n}x^{\frac{1}{n}-1} \\&= \frac{1}{n}x^{r-1}.\end{aligned}$$

6 Indefinite Integral

Whenever a certain operation is considered in mathematics, the question arises of the operation inverse to it. For example, alongside the operation of addition, we consider its inverse subtraction. Alongside multiplication, we consider division; alongside exponentiation, we consider the inverse operation of extracting roots.

When examining an inverse operation, two fundamental questions arise: the possibility of performing the operation, and its uniqueness. For instance, if we restrict ourselves to real numbers, extracting the square root is not always possible — we cannot extract the square root of a negative number. Moreover, extracting a square root is not a one-to-one operation, since extracting the square root of a positive number yields two values: one positive and one negative.

Now that we have introduced the operation of differentiation, the question arises of the operation inverse to it, which is called *integration*. For the operation of integration, we must answer two main questions: the feasibility of integration, and the uniqueness of the integral.

Let us proceed to exact mathematical formulations. Suppose a function $f(x)$ is given. A function $h(x)$, defined for the same values of the argument as $f(x)$, and satisfying the condition

$$h'(x) = f(x), \quad (6.1)$$

is called the integral of the function $f(x)$ or a *primitive* (also: *antiderivative*) of the function $f(x)$. The transition from a given function $f(x)$ to a function $h(x)$ satisfying equation (6.1) is the operation inverse to differentiation, and is called *integration*.

It is immediately clear that the integration operation is not unique. Indeed, if a function $h(x)$ satisfies equation (6.1), then the function $h(x) + c$, where c is a constant, also satisfies the same equation. In fact, we have

$$(h(x) + c)' = h'(x) + c' = h'(x) + 0 = f(x). \quad (6.2)$$

(see Section 2, (2.14)). It turns out, however, that this is the only source of ambiguity in the integration op-

eration: all primitives of a given function differ by a constant. Let us prove this.

First, let us prove that if a function $h(x)$ satisfies the equation

$$h'(x) = 0, \quad (6.3)$$

then the function $h(x)$ is constant:

$$h(x) = c. \quad (6.4)$$

This statement is true, however, only when the domain of definition of the function $h(x)$ is *connected*. Connectedness means that, along with any two values x_1 and x_2 of the argument for which $h(x)$ is defined, every number intermediate between x_1 and x_2 must also be included in the domain of $h(x)$. One must never forget this.

Let us now prove that equation (6.3) implies equation (6.4). Let x_1 and x_2 be two arbitrary argument values of the function $h(x)$. Since the domain of $h(x)$ is connected, it is defined on the whole interval $[x_1, x_2]$. Therefore, by the Lagrange Mean Value Theorem (see Section 3, (3.14)), we have

$$h(x_2) - h(x_1) = h'(\theta)(x_2 - x_1). \quad (6.5)$$

Since $h'(\theta) = 0$ (see (6.3)), it follows from equation (6.5), proving equation (6.4). Now let $h_1(x)$ and $h_2(x)$ be two

functions satisfying the equations

$$h_1'(x) = f(x), \quad h_2'(x) = f(x), \quad (6.6)$$

so that $h_1(x)$ and $h_2(x)$ are both primitives of the same function $f(x)$. We shall prove that

$$h_2(x) = h_1(x) + c, \quad (6.7)$$

where c is a constant. Indeed, let us set

$$\begin{aligned} h'(x) &= (h_2(x) - h_1(x))' = h_2'(x) - h_1'(x) \\ &= f(x) - f(x) = 0, \end{aligned}$$

so that $h(x)$ is a *constant function*. From this it follows that equation (6.7) holds.

In the proof we used the fact that the functions $h_1(x)$, $h_2(x)$, and therefore also $f(x)$, are defined on a connected set. The solution $h(x)$ of equation (6.1) is written in the form

$$h(x) = \int f(x) \, dx, \quad (6.8)$$

The symbol $\int$ in formula (6.8) is read as *integral*. Since this formula defines the function $h(x)$ only up to an *additive constant*, the integral written on the right-hand side of the formula is called an *indefinite integral*.

The question of finding the solution $h(x)$ of equation (6.1) for a given function $f(x)$ can, at present, be

answered affirmatively for certain specific functions $f(x)$. The method of solution practically reduces to guessing the solution based on the formulas provided in Section 2 (2.2)–(2.5). For example, if the function $f(x)$ is a polynomial

$$f(x) = c_0x^n + c_1x^{n-1} + \dots + c_{n-1}x + c_n, \quad (6.9)$$

then using formula (2.18) of Section 2, we can find the function $h(x)$. Namely, we have

$$\begin{aligned} h(x) &= \int (c_0x^n + c_1x^{n-1} + \dots + c_{n-1}x + c_n) dx \\ &= \frac{c_0}{n+1} x^{n+1} + \frac{c_1}{n} x^n + \dots + \frac{c_{n-1}}{2} x^2 + c_n x + c, \end{aligned} \quad (6.10)$$

where c is an arbitrary constant.

In the same way, from formulas (5.8) and (5.7) of Section 5, we obtain:

$$\begin{aligned} \int \sin x \, dx &= -\cos x + c, \\ \int \cos x \, dx &= \sin x + c, \end{aligned} \quad (6.11)$$

where c is an arbitrary constant.

From formula (6.11) of Section 5, we also have

$$\int \frac{dx}{\cos^2 x} = \tan x + c. \quad (6.12)$$

There exist many different methods for finding indefinite integrals, but all of them are, in practice, reduced to guessing. We will not present them here. Instead, we will give one general rule.

If for the functions $f_1(x)$, $f_2(x)$, $\dots$, $f_m(x)$ their antiderivatives $h_1(x)$, $h_2(x)$, $\dots$, $h_m(x)$ are known, so that the relations $h'_i(x) = f_i(x)$, $i = 1, 2, \dots, m$ hold, then for the function

$$f(x) = c_1 f_1(x) + c_2 f_2(x) + \dots + c_m f_m(x), \quad (6.13)$$

where $c_1, c_2, \dots, c_m$ are constants, we can write the indefinite integral as

$$\int f(x) dx = c_1 h_1(x) + c_2 h_2(x) + \dots + c_m h_m(x) + c, \quad (6.14)$$

where c is an arbitrary constant.

Let us now give a simple but very important application of these results to a problem in mechanics.

Let us consider the motion of a point along a straight line. To describe this motion mathematically, we take the line as the x -axis, and denote the position of the point at time t by $x(t)$, understanding $x(t)$ to represent both the point itself and its coordinate.

The function $x(t)$, which depends on time t , completely describes the motion of the point as a function

of time. If t and τ are two moments in time such that $t < \tau$, then over the time interval $\tau - t$, the point travels the distance $x(\tau) - x(t)$, and the average velocity of the point over the time interval $[t, \tau]$ is naturally given by the formula

$$\frac{x(\tau) - x(t)}{\tau - t}. \quad (6.15)$$

The closer τ is to t , the more accurately the expression (6.15) represents the instantaneous velocity at time t . Thus, the instantaneous velocity $v(t)$ is defined by the formula

$$v(t) = \lim_{\tau \rightarrow t} \frac{x(\tau) - x(t)}{\tau - t}. \quad (6.16)$$

The right-hand side of this equation is nothing other than the derivative of the function $x(t)$ with respect to t , so that the velocity $v(t)$ of the point at time t is given by the expression

$$v(t) = x'(t). \quad (6.17)$$

If the velocity $v(t)$ of the moving point does not depend on t , i.e., the point moves with constant velocity $v(t) = v$, where v is a constant value, then the position of the point $x(t)$ should be found from the differential equation

$$x'(t) = v. \quad (6.18)$$

According to formula (6.10), the solution of this equation is written as

$$x(t) = vt + c, \quad (6.19)$$

where c is a constant. To determine this constant, let x_0 be the position of the point $x(t)$ at time $t = 0$. Substituting $x(0) = x_0$ into equation (6.19), we obtain

$$x(0) = x_0 = c. \quad (6.20)$$

Thus, the solution of equation (6.18) can be written as

$$x(t) = x_0 + vt, \quad (6.21)$$

where x_0 is the position of the point at the initial time $t = 0$, and v is the constant velocity of the point. Equation (6.21) describes the motion of a point with constant velocity v .

If the velocity of the point is not constant, then in addition to velocity, another important quantity is considered *acceleration*, which characterises the change in velocity. Analogous to how we defined the average velocity, we define the *average acceleration* over the time interval from t to τ . It is given by the formula

$$\frac{v(\tau) - v(t)}{\tau - t}. \quad (6.22)$$

The closer the moment τ is to the moment t , the more accurately formula (6.22) represents the acceleration at time t . Thus, the exact value of the acceleration $a(t)$ of the point at time t is given by the formula

$$a(t) = \lim_{\tau \rightarrow t} \frac{v(\tau) - v(t)}{\tau - t}. \quad (6.23)$$

The right-hand side of this equality is nothing other than the derivative $v'(t)$ of the function $v(t)$ with respect to the independent variable t , so we have

$$a(t) = v'(t). \quad (6.24)$$

Taking into account formula (6.17), we can rewrite equation (6.24) in the form

$$a(t) = x''(t), \quad (6.25)$$

(see Section 4, (4.1)). Thus, the *velocity* of the point $x(t)$ at time t is the *first derivative* $x'(t)$, and the *acceleration* of the point $x(t)$ at time t is the *second derivative* $x''(t)$.

Uniformly accelerated motion, that is, motion in which the acceleration $a(t)$ is a constant quantity, is of great interest. In this case,

$$a(t) = a, \quad (6.26)$$

where a is a constant. Then the velocity $v(t)$ of the moving point is obtained from the differential equation

$$v'(t) = a. \quad (6.27)$$

By formula (6.10), the solution of this equation is

$$v(t) = at + c, \quad (6.28)$$

where c is a constant. To find this constant, let v_0 be the velocity of the point $x(t)$ at time $t = 0$. Substituting $t = 0$ into equation (6.28), we obtain

$$v(0) = v_0 = c.$$

Thus, the solution of equation (6.27) is

$$v(t) = v_0 + at.$$

Substituting $v(t)$ into the expression for $x'(t)$, we obtain the equation

$$x'(t) = v_0 + at. \quad (6.29)$$

According to the previously established results (see formula (6.10)), the solution of this equation is the function

$$x(t) = v_0 t + \frac{at^2}{2} + c, \quad (6.30)$$

where c is a constant. To determine this constant, let x_0 be the position of the point $x(t)$ at time $t = 0$. Substituting $t = 0$ into equation (6.30), we find

$$x(0) = x_0 = c.$$

Thus, the equation of uniformly accelerated motion is

$$x(t) = x_0 + v_0 t + \frac{at^2}{2}. \quad (6.31)$$

7 Definite Integral

Integration arises in mathematics not only as an operation that is the reverse of differentiation, but also in solving many other problems, in particular when calculating areas in geometry. The calculation of the area of figures on a plane, bounded by curved lines, leads us to integration. In this case, the integral ceases to be indefinite, since the area of a figure has a well-defined value. We will consider here the simplest problem of calculating the area, which leads to a definite integral.

Let us start with a given function

$$y = f(x), \quad (7.1)$$

and the graph L of this function, plotted in the usual rectangular Cartesian coordinate system. Let us assume for now that the entire graph L of $f(x)$ lies above the

x -axis. Let u and v be two arbitrary values of the argument x of the function $f(x)$, where $u < v$. We will denote the points on the graph of $f(x)$ with x -coordinates u and v by a and b , respectively.

Now, naturally, a portion of the plane is delineated, bounded from below by the segment $[u, v]$ of the x -axis, on the left by the ordinate at a , on the right by the ordinate at b , and above by the segment $[a, b]$ of the graph L of $f(x)$. This portion of the plane is denoted as $Q(u, v)$. Common sense and the needs of application tell us that the region $Q(u, v)$ has a definite area.

We will denote this area by $h(u, v)$. Now, let us choose two values of the argument x_0 and x of the function $f(x)$, and set

$$h(x_0, x) = h(x). \quad (7.2)$$

Denoting the corresponding area by $h(x)$, we thereby emphasise that we are considering it as a function of the variable x . We now compute the derivative $h'(x)$ of the function $h(x)$ defined in this way.

Let us choose some value ξ of the argument of our function (7.1), close to x . For definiteness, we assume $\xi > x$. To compute the derivative $h'(x)$, we must first compute the difference $h(\xi) - h(x)$, that is, the difference in the areas of the figures $Q(x_0, \xi)$ and $Q(x_0, x)$.

This difference is equal to the area of the figure $Q(x, \xi)$. We have

$$h(\xi) - h(x) = h(x, \xi) \quad (7.3)$$

We shall not compute the area $h(x, \xi)$ of the figure $Q(x, \xi)$ exactly, but only give an estimate. For this purpose, we introduce new notations. For two arbitrary values u and v of the argument of the function $f(x)$, we denote by $\mu(u, v)$ the minimum of the function $f(x)$ on the segment $[u, v]$, and by $\nu(u, v)$ the maximum of the function $f(x)$ on the same segment $[u, v]$.

The rectangle with base $[u, v]$ and height $\mu(u, v)$ will be denoted by $M(u, v)$, and the rectangle with base $[u, v]$ and height $\nu(u, v)$ will be denoted by $N(u, v)$. Applying this notation to the case where $u = x$, $v = \xi$, we see that the rectangle $M(x, \xi)$ is contained within the figure $Q(x, \xi)$, and the figure $Q(x, \xi)$ is itself contained within the rectangle $N(x, \xi)$.

Since the areas of the rectangles $M(x, \xi)$ and $N(x, \xi)$ are respectively

$$(\xi - x) \cdot \mu(x, \xi), \quad (\xi - x) \cdot \nu(x, \xi) \quad (7.4)$$

Then we obtain the following double inequality:

$$(\xi - x) \cdot \mu(x, \xi) \leq h(\xi) - h(x) \leq (\xi - x) \cdot \nu(x, \xi) \quad (7.5)$$

or, in another form,

$$(\xi - x) \cdot \mu(x, \xi) \leq h(\xi) - h(x) \leq (\xi - x) \cdot \nu(x, \xi) \quad (7.6)$$

Clearly, as $\xi \rightarrow x$, we have

$$\mu(x, \xi) \rightarrow f(x), \quad \nu(x, \xi) \rightarrow f(x). \quad (7.7)$$

Thus, dividing the double inequality (7.6) by the positive quantity $(\xi - x)$ and passing to the limit as $\xi \rightarrow x$, we obtain

$$f(x) \leq \lim_{\xi \rightarrow x} \frac{h(\xi) - h(x)}{\xi - x} \leq f(x). \quad (7.8)$$

Since, by definition,

$$h'(x) = \lim_{\xi \rightarrow x} \frac{h(\xi) - h(x)}{\xi - x},$$

it follows from formula (7.8) that

$$h'(x) = f(x). \quad (7.9)$$

Thus, we have found the derivative of the function $h(x)$ and have shown that $h(x)$ is a primitive (antiderivative) of the function $f(x)$. We also note one more property of the function $h(x)$: when $x = x_0$, the portion of the plane $Q(x_0, x)$ degenerates into a line segment and therefore has area zero. Thus,

$$h(x_0) = 0. \quad (7.10)$$

All that has been done so far was based on the assumption that $x_0 \leq x$ and that the graph L of the function

$f(x)$ on the interval $[x_0, x]$ lies entirely above the x -axis. These assumptions must now be lifted, and the function $h(x)$ must be defined in such a way that it satisfies conditions (7.9) and (7.10).

If the graph of the function $f(x)$ on the interval $[x_0, x]$ lies partly above and partly below the x -axis, we divide it into two parts: L_+ , which lies above the x -axis, and L_- , which lies below the x -axis. Each of the curves L_+ and L_- may consist of several segments. We denote the area enclosed between the curve L_+ and the x -axis by $h_+(x)$, and the area between the x -axis and the curve L_- by $h_-(x)$. Retaining the assumption that $x_0 < x$, we now define

$$h(x) = h_+(x) - h_-(x) \quad (7.11)$$

If now $x_0 > x$, then the areas $h_+(x)$ and $h_-(x)$ are defined in exactly the same way as in the case $x_0 < x$, but the function $h(x)$ is defined by the formula

$$h(x) = -(h_+(x) - h_-(x)) \quad (7.12)$$

Area, by its very nature, is a positive quantity. By defining it through formulas (7.11) and (7.12), we thereby algebraise the area, that is, we assign it a positive or negative sign depending on the circumstances. Perhaps somewhat laboriously, but without any essential difficulties, it can be proven that under the new definition

of the function $h(x)$, its properties (7.9) and (7.10) are preserved. Thus, for any function $f(x)$, a corresponding antiderivative $h(x)$ has been found, satisfying the additional condition (7.10).

This method of constructing the function $h(x)$, which is an antiderivative of the function $f(x)$, does convince us that the function $h(x)$ exists, since by common sense the area of a figure exists, but it does not provide us with a way to compute this area, for example, with the help of an electronic computing machine. A method for calculating the function $h(x)$ will be given somewhat later, but for now we shall demonstrate that the result already obtained can still yield certain outcomes.

It allows us to compute the area $h(x)$ of the piece of plane $Q(x_0, x)$ in the case when we are able to guess some antiderivative $h_1(x)$ of the function $f(x)$. So, let us suppose that we have somehow found a function $h_1(x)$ satisfying the condition

$$h_1'(x) = f(x). \quad (7.13)$$

Substituting into equation (6.7) in Section 6 the function $h(x)$ instead of $h_1(x)$, we obtain the equality

$$h(x) = h_1(x) + c \quad (7.14)$$

where c is a constant. To determine it, we substitute $x = x_0$ into this equation. Then, using equation (7.10),

7 Definite Integral

we obtain

$$0 = h(x_0) = h_1(x_0) + c.$$

Hence, it follows that $c = -h_1(x_0)$, and therefore

$$h(x) = h_1(x) - h_1(x_0). \quad (7.15)$$

This formula represents an important result. It expresses the area $h(x)$ of the figure $Q(x_0, x)$ through an arbitrary antiderivative $h_1(x)$ of the function $f(x)$. The function $h(x)$ itself is called the *definite integral* of the function $f(x)$ and is written as

$$h(x) = \int_{x_0}^x f(t) \, dt. \quad (7.16)$$

Here, x_0 is called the *lower limit* of integration, x the *upper limit* of integration, and t the *variable of integration*. The function $h(x)$ depends on the given function $f(x)$ and the limits of integration x_0 and x , but not at all on the variable of integration itself, or more precisely, on its symbol. Thus, it can be denoted by any letter. For example, we can write

$$h(x) = \int_{x_0}^x f(\tau) \, d\tau.$$

Definite Integral as the Limit of a Sequence of Finite Sums

Let us now move on to the method of approximate calculation of the area $h(x)$ of the region $Q(x_0, x)$. Here again, we shall assume that $x_0 < x$ and that on the interval $[x_0, x]$, the graph of the function $f(x)$ lies above the x -axis. Let us divide the interval $[x_0, x]$ by points

$$x_0 < x_1 < x_2 < \dots < x_n = x. \quad (7.17)$$

We shall say that the subdivision (7.17) of the interval $[x_0, x]$ has *fineness* δ if the length of each subinterval is less than δ , that is, if the condition

$$x_i - x_{i-1} < \delta \quad (i = 1, 2, \dots, n). \quad (7.18)$$

is satisfied. Let us use the previously introduced notation for the interval $[u, v]$, now applied to the subinterval $[x_{i-1}, x_i]$, and compose two sums:

$$\begin{aligned} M = & (x_1 - x_0)\mu(x_0, x_1) + (x_2 - x_1)\mu(x_1, x_2) + \\ & \dots + (x_n - x_{n-1})\mu(x_{n-1}, x_n), \end{aligned} \quad (7.19)$$

$$\begin{aligned} N = & (x_1 - x_0)\nu(x_0, x_1) + (x_2 - x_1)\nu(x_1, x_2) + \\ & \dots + (x_n - x_{n-1})\nu(x_{n-1}, x_n). \end{aligned} \quad (7.20)$$

The sum (7.19) represents the total area of all rectangles

$$M(x_{i-1}, x_i) \quad (i = 1, 2, \dots, n), \quad (7.21)$$

while the sum (7.20) represents the total area of all rectangles

$$N(x_{i-1}, x_i) \quad (i = 1, 2, \dots, n). \quad (7.22)$$

The union of all rectangles in (7.21) lies within the region of the plane $Q(x_0, x)$. The union of all rectangles in (7.22) contains within it the region $Q(x_0, x)$. Thus, we obtain the double inequality

$$M \leq h(x) \leq N. \quad (7.23)$$

The values M and N depend on the sequence of points in (7.17), i.e., on the way the interval $[x_0, x]$ is subdivided. It can be shown — somewhat tediously but without serious difficulty — that when the subdivision (7.17) is refined indefinitely, i.e., as $\delta \rightarrow 0$, the quantity $N - M$ also tends to zero. This shows that the quantities M and N are approximate values of the area $h(x)$.

A certain inconvenience in calculating M and N is that one has to find the maxima and minima of the function $f(x)$ on all the subintervals. This can be simplified as follows: on each interval $[x_{i-1}, x_i]$, choose an arbitrary point ξ_i , and form the sum

$$\begin{aligned} P = & (x_1 - x_0)f(\xi_1) + (x_2 - x_1)f(\xi_2) + \\ & \dots + (x_n - x_{n-1})f(\xi_n). \end{aligned} \quad (7.24)$$

Since the obvious inequality holds:

$$\mu(x_{i-1}, x_i) \leq f(\xi_i) \leq \nu(x_{i-1}, x_i), \quad (7.25)$$

we obtain the inequality

$$M \leq P \leq N. \quad (7.26)$$

Since the values M and N converge indefinitely as $\delta \rightarrow 0$, the values $h(x)$ and P also converge indefinitely as the subdivision (7.17) is infinitely refined. Thus, the area $h(x)$ can be approximately calculated using the sum (7.24). This is the method of computing the area of the region $Q(x_0, x)$, and it may also be regarded as a logical definition of the concept of area.

8 Postulate of Convergence

We shall now give a very simple example of computing area using formula (7.16), and in connection with this discussion, introduce a very important criterion for the existence of a limit. Let us define the function by the formula

$$f(x) = \frac{1}{x^2} = x^{-2} \quad \text{for } x > 0 \quad (8.1)$$

According to formula (5.23) from Section 5, the antiderivative of the function x^{-2} is the function $-x^{-1}$. Indeed,

$$\left(-x^{-1}\right)' = x^{-2} = \frac{1}{x^2}.$$

Thus, according to formula (7.16), the area of the region $Q(x_0, x)$, defined by the function (8.1), is given by

$$h(x) = \int_{x_0}^x \frac{dt}{t^2} = -\left(\frac{1}{x} - \frac{1}{x_0}\right) = \frac{1}{x_0} - \frac{1}{x}.$$

Hence, we have

$$h(x) = \frac{1}{x_0} - \frac{1}{x}. \quad (8.2)$$

Since $x > x_0$, it follows that $h(x) > 0$. Here it is important to note one very interesting fact. If x increases without bound, then the right-hand side of equation (8.2) tends to the limit $1/x_0$. This is to be understood as meaning that there is a finite area between the ordinate x_0 , the x -axis, and the curve

$$y = \frac{1}{x^2},$$

even though the strip enclosed between the curve and the x -axis extends to infinity. This is usually written in the form

$$\int_{x_0}^{\infty} \frac{dt}{t^2} = \frac{1}{x_0}. \quad (8.3)$$

This somewhat surprising phenomenon is connected with another phenomenon which plays a very important role. We shall now turn our attention to it.

Let us suppose that k is a natural number and construct the sequence of integers

$$x_0 = k, \quad x_1 = k+1, \dots, x_i = k+i, \quad \dots, \quad x_n = k+n = x. \quad (8.4)$$

This sequence of integers divides the interval of integration $[x_0, x]$ into subintervals of length 1. Let us construct, for this subdivision of the interval $[x_0, x]$ into subintervals of length 1, the sum M (see equation (7.19)) for the function (8.1). In this case, we must take into account that the minimum of the function $f(x)$ on the interval $[x_{i-1}, x_i]$ is achieved at the right end of the interval and equals $1/x_i^2$:

$$\mu(x_{i-1}, x_i) = \frac{1}{x_i^2} = \frac{1}{(k+i)^2}. \quad (8.5)$$

The number M in our case depends on the yet unspecified natural numbers k and n , so we denote it by $M(k, n)$. By formula (7.19), we obtain for it the expression:

$$M = M(k, n) = \frac{1}{(k+1)^2} + \frac{1}{(k+2)^2} + \dots + \frac{1}{(k+n)^2}. \quad (8.6)$$

From the first part of the double inequality (7.23), we get in our case:

$$M(k, n) \leq \int_k^{k+n} \frac{dt}{t^2} = \frac{1}{k} - \frac{1}{k+n}. \quad (8.7)$$

Now let us set

$$k = 1, \quad p = n + 1, \quad s_p = M(1, p). \quad (8.8)$$

Then we obtain the following expression for s_p :

$$\begin{aligned} s_p &= \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \\ \dots + \frac{1}{p^2} &\leq 1 + 1 - \frac{1}{p} = 2 - \frac{1}{p}. \end{aligned} \quad (8.9)$$

It follows from this that the quantity s_p always satisfies the inequality:

$$s_p \leq 2. \quad (8.10)$$

From equation (8.9), it is clear that the quantity s_p increases as p increases, but remains bounded during this growth, always staying less than 2. Common sense tells us that the quantity s_p , increasing without bound in p yet remaining bounded, must approach some definite number s . This assertion can be written as:

$$\lim_{p \rightarrow \infty} s_p = s. \quad (8.11)$$

Postulate of Convergence

We shall accept without proof the following assertion.

Suppose there is a certain function $\sigma(p)$ of the natural number p , satisfying two conditions: the function $\sigma(p)$ increases with increasing p , i.e., the condition holds:

$$\sigma(p+1) > \sigma(p) \quad (p = 1, 2, \dots) \quad (8.12)$$

and $\sigma(p)$ remains bounded for all values of p , i.e.,

$$\sigma(p) < c, \quad (8.13)$$

where c is a constant not depending on p . Then the quantity $\sigma(p)$, as p increases without bound, gets arbitrarily close to some number σ . This assertion is written as

$$\lim_{p \rightarrow \infty} \sigma(p) = \sigma. \quad (8.14)$$

We can now formulate relation (8.9) in a different way. Let us consider the infinite sum

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{p^2} + \dots \quad (8.15)$$

Since the partial sum s_p (see equation (8.9)) of the terms of this series satisfies the conditions of our postulate, we shall assume that the infinite sum (8.15) exists and is given by the formula

$$s = \lim_{p \rightarrow \infty} s_p. \quad (8.16)$$

It is said that the infinite series (8.15) converges and has the sum s .

If

$$p < q, \quad (8.17)$$

— where p and q are two natural numbers — then from formulas (8.6), (8.7) and (8.9) it follows that

$$s_q = s_p + M(p, q - p), \quad (8.18)$$

where

$$M(p, q - p) \leq \frac{1}{p}. \quad (8.19)$$

Thus, we have

$$s_q - s_p \leq \frac{1}{p}. \quad (8.20)$$

Taking the limit as $q \rightarrow \infty$ in this relation, we obtain

$$s - s_q \leq \frac{1}{p}. \quad (8.21)$$

from which we see that the value s is determined by the value s_p with an accuracy up to $1/p$. Therefore, when we consider the value s_p as an approximate value of s , we know the error committed in this approximation.

In algebra, we have already encountered the sum of an infinite series – namely, the sum of an infinite geometric progression:

$$s = w + wv + wv^2 + \dots + wv^i + \dots$$

In the case where $|v| < 1$, this progression has a sum equal to

$$\frac{w}{1 - v}. \quad (8.22)$$

Thus, we have already dealt with summing an infinite series and had the opportunity to compute its sum as in (8.22).

The infinite series (8.15) also has a sum, but we cannot express it using a formula as we did for the geometric progression. We can only be certain that the finite sum s_p , composed of its initial terms, is an approximate value for the sum s of this series, and we know the accuracy of this approximation (see (8.21)).

Since we can find an approximation of the number s with any desired degree of accuracy, we must consider that the number s is known to us. We have already encountered such a phenomenon when discussing the number π . It can be calculated with any desired accuracy, but it cannot be represented by any algebraic formula.

The same applies to the number s , which is defined by equation (8.16). The numbers s_p are its approximate values.

9 Newton's Binomial and the Sum of a Geometric Progression

Here we shall prove the algebraic formulas mentioned in the title of this section, which will be needed in Section 10.

Newton's Binomial

To state and prove this formula of Newton's, known as Newton's binomial, we must first recall what is meant by the function $n!$ of a natural number n . This function

is defined by the formula

$$n! = 1 \cdot 2 \cdot 3 \cdots n. \quad (9.1)$$

Thus, $n!$ (read: “ n factorial”) is the product of all natural numbers in succession from 1 to n . We have

$$\begin{aligned} 1! &= 1, & 2! &= 1 \cdot 2 = 2, & 3! &= 1 \cdot 2 \cdot 3 = 6, \\ 4! &= 1 \cdot 2 \cdot 3 \cdot 4 = 24. \end{aligned}$$

For convenience, we also define

$$0! = 1. \quad (9.2)$$

The Newton formula in question is written as follows:

$$(a + b)^n = \sum_{\substack{i, j \\ i+j=n}} \frac{n!}{i! j!} a^i b^j. \quad (9.3)$$

On the right-hand side stands the sum of all terms of the form

$$\frac{n!}{i! j!} a^i b^j, \quad (9.4)$$

where $i + j = n$, and i and j are non-negative integers. We shall prove this formula by induction. That is, we shall first verify that it holds for $n = 1$, and then prove that if it holds for an exponent equal to n , it also holds for the exponent equal to $n + 1$.

For $n = 1$, we have

$$(a + b)^1 = a + b = \frac{a}{1!0!} + \frac{b}{0!1!}. \quad (9.5)$$

Thus, for $n = 1$, formula (9.3) holds.

To carry out the induction step, we multiply the identity (9.3) by $a + b$. On the left-hand side, we get $(a + b)^{n+1}$. On the right-hand side, we obtain terms containing the factor $a^r b^s$, where $r + s = n + 1$.

From term (9.4), after multiplying by $(a + b)$, we obtain a term containing the factor $a^r b^s$ in the following cases:

$$i = r1, \quad j = s, \quad \text{or} \quad i = r, \quad j = s1.$$

In the first case, we obtain a term containing $a^r b^s$ by multiplying term (9.4) by u , while in the second case, it arises from multiplying term (9.4) by v .

Thus, after multiplying the sum on the right-hand side of equality (9.3) by $(a + b)$, we arrive at a term containing $a^r b^s$ with the coefficient:

$$\frac{n!}{(r-1)!s!} = \frac{n!}{r!(s-1)!}. \quad (9.6)$$

Grouping all terms with the same powers $a^r b^s$, where $r + s = n + 1$, we find that the coefficient of such a term

is:

$$\frac{n!}{(r-1)!s!} + \frac{n!}{r!(s-1)!} = \frac{(n+1)!}{r!s!}.$$

By multiplying the numerator and denominator of the first term in this sum by a , and the numerator and denominator of the second term by b , we rewrite sum (9.6) in the form:

$$\frac{n!r}{r!s!} + \frac{n!s}{r!s!} = \frac{n!(r+s)}{r!s!} = \frac{(n+1)!}{r!s!}. \quad (9.7)$$

Thus, we finally obtain: Thus, we obtain the identity:

$$(a+b)^{n+1} = \sum_{\substack{r,s \\ r+s=n+1}} \frac{(n+1)!}{r!s!} a^r b^s. \quad (9.8)$$

Hence, the inductive step is complete, and formula (9.3) is proved.

$$\frac{n!}{i!j!} \quad (9.9)$$

The coefficient appearing in formula (9.3) is usually written in a slightly different form. For this, we set $j = k$, $i = n - k$, and simplify the fraction (9.3) by cancelling out $(n - k)!$. Indeed, we have:

$$\frac{n!}{(n-k)!} = n(n-1)\dots(n-k+1). \quad (9.10)$$

Thus, we obtain:

$$\frac{n!}{(n-k)!k!} = \frac{n(n-1)\dots(n-k+1)}{k!}, \quad (9.11)$$

and formula (9.3) is rewritten as:

$$\begin{aligned} (a+b)^n &= a^n + \frac{n}{1!}a^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \dots \\ &+ \frac{n(n-1)\dots(n-k+1)}{k!}a^{n-k}b^k + \dots \\ &+ b^n. \end{aligned} \quad (9.12)$$

Sum of a Geometric Progression

Let us denote:

$$g_m = w + w \cdot v + w \cdot v^2 + \dots + w \cdot v^m. \quad (9.13)$$

Multiply and divide the right-hand side of this expression by $(1-v)$, and use formula (2.9), setting $u = 1$. Then we obtain:

$$g_m = w \frac{(1-v^{m+1})}{1-v}. \quad (9.14)$$

In what follows, we shall use this formula only in the case:

$$0 < v < 1, \quad w > 0.$$

In this case, it follows that:

$$g_m < \frac{w}{1-v}. \quad (9.15)$$

10 The Function e^x

Here we shall first define the function e^x in a very pedantic and rigorous manner, where x is a variable that may take arbitrary real values, and e is a well-known and important number in mathematics:

$$e = 2.71828\dots$$

We begin our investigation with the function

$$\omega_n(x) = \left(1 + \frac{x}{n}\right)^n. \quad (10.1)$$

where n is a natural number. The function $\omega_n(x)$ defined by this expression is a polynomial of degree n in x . First, we shall prove that for every fixed x , the value of $\omega_n(x)$ tends to a certain definite limit as $n \rightarrow \infty$, which we denote by $\exp(x)$. In formula form, this is

written:

$$\lim_{n \rightarrow \infty} \omega_n(x) = \exp(x). \quad (10.2)$$

As a result of a careful study of the function $\exp(x)$, defined by this relation, we shall arrive at the conclusion that:

$$\exp(x) = e^x \quad (10.3)$$

Study of the Function $\omega_n(x)$

We apply formula (9.12) to the right-hand side of equation (10.1), setting $u = 1$, $v = x/n$. Then we obtain:

$$\begin{aligned} \omega_n(x) &= 1 + \frac{n}{1!} \frac{x}{n} + \dots \\ &+ \frac{n(n-1) \dots (n-k+1)}{k!} \frac{x^k}{n^k} + \dots \end{aligned} \quad (10.4)$$

We transform the numerical coefficient in front of x^k . We have:

$$= 1 \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{k-1}{n}\right) \frac{1}{k!}. \quad (10.5)$$

Setting:

$$\gamma_n^{(k)} = 1 \left(1 - \frac{1}{n}\right) \dots \left(1 - \frac{k-1}{n}\right), \quad (10.6)$$

We can now rewrite formula (10.4) as:

$$\omega_n(x) = 1 + \gamma_n \frac{x}{1!} + \dots + \gamma_n(k) \frac{x^k}{k!} + \dots \quad (10.7)$$

The quantity $\gamma_n^{(k)}$ has the following properties:

$$y_n^{(1)} = 1 \quad (10.8)$$

$$0 < y_n^{(k)} < y_{n+1}^{(k)} < 1 \quad \text{for } k \leq n \quad (10.9)$$

$$y_n^{(k)} = 0 \quad \text{for } k > n$$

From formula (10.7) and inequality (10.9), it follows that for $x > 0$, we have:

$$\omega_n(x) < \omega_{n+1}(x), \text{ when } x > 0. \quad (10.10)$$

Now let us fix a value of x , not necessarily positive, and choose a sufficiently large natural number p such that: $|x| < p + 1$ or equivalently

$$\frac{|x|}{k} < 1. \quad (10.11)$$

Then for $k > p$, we have:

$$\begin{aligned} \frac{\gamma_n(k)}{k!} |x|^k &\leq \frac{|x|^k}{k!} = \frac{|x|^p}{p!} \cdot \frac{|x|}{p+1} \cdot \frac{|x|}{p+2} \cdot \frac{|x|}{k} \\ &\leq \frac{|x|^p}{p!} \cdot \frac{|x|^{k-p}}{(p+1)^{k-p}}. \end{aligned} \quad (10.12)$$

We split the sum in equation (10.7) into two parts:

$$\omega_n(x) = s_n(p, x) + r_n(p, x), \quad (10.13)$$

where

$$s_n(p, x) = 1 + \gamma_n^{(1)} \cdot \frac{x}{1!} + \dots + y_n^{(i)} \cdot \frac{x^i}{i!} + \dots + y_n^{(p)} \cdot \frac{x^p}{p!} \quad \text{and} \quad (10.14)$$

$$r_n(p, x) = \gamma_n^{(p+1)} \cdot \frac{x^{p+1}}{(p+1)!} + \gamma_n^{(k)} \cdot \frac{x^k}{(k)!} + \dots \quad (10.15)$$

Here, $i \leq p$, and $k > p$.

By equation (10.12) the inequality:

$$\begin{aligned} |r_n(p, x)| &\leq \frac{|x|^p}{p!} \cdot \left[\frac{|x|}{p+1} + \frac{|x|^2}{(p+1)^2} + \dots + \frac{|x|^{k-p}}{(p+1)^{k-p}} \right] \\ &< \frac{|x|^p}{p!} \cdot \frac{\frac{|x|}{p+1}}{1 - \frac{|x|}{p+1}} = \frac{|x|^p}{p!} \cdot \frac{|x|}{p+1-|x|}. \end{aligned}$$

See also equation (9.15) and inequality (10.11), we finally obtain

$$|r_n(p, x)| = \frac{|x|^p}{p!} \cdot \frac{|x|}{p+1-|x|} \quad \text{for } |x| < p+1. \quad (10.16)$$

From this, it is clear that the quantity $|r_n(p, x)|$ remains bounded as $n \rightarrow \infty$, and hence we have the inequality:

$$\omega_n(x) < c(x), \quad (10.17)$$

where $c(x)$ depends on x , but not on n . Thus, for $x > 0$, by inequalities (10.10) and (10.17), the quantity $\omega_n(x)$ increases with increasing n , while remaining bounded. Therefore, its limit exists (see Section 8), so we can write:

$$\exp(x) = \lim_{n \rightarrow \infty} \omega_n(x), \text{ for } x > 0.$$

Case $|x| \leq 1$

In this case, we may assume $p = 1$, since when $|x| < 1$, inequality (10.11) is satisfied for $p = 1$. Then formula (10.13) becomes:

$$\omega_n(x) = 1 + x + r_n(1, x), \text{ where}$$

$$|r_n(1, x)| < |x| \cdot \frac{|x|}{2 - |x|} \leq |x|^2.$$

Thus, finally, when $|x| \leq 1$, we obtain

$$\omega_n(x) = 1 + x + r_n(1, x), \text{ with } |r_n(1, x)| < x^2. \quad (10.18)$$

Now let us consider a sequence

$$\xi_1, \xi_2, \dots, \xi_n, \dots \quad (10.19)$$

i.e., ω_n is some sequence converging to zero. Consider the quantity $\omega_n(\xi_n)$. Since for sufficiently large n , we have $|\xi_n| < 1$, then by formula (10.18) we obtain:

$$\begin{aligned}\omega_n(\xi_n) &= 1 + \xi_n + r_n(1, \xi_n), \text{ where} \\ |r_n(1, \xi_n)| &< \xi_n^2.\end{aligned}$$

From this it follows that:

$$\text{If } \lim_{n \rightarrow \infty} \xi_n = 0, \text{ then } \lim_{n \rightarrow \infty} \omega_n(\xi_n) = 1. \quad (10.20)$$

Now let us consider the case when the argument x of the function $\omega_n(x)$ is negative. For this purpose, we consider the function

$$\omega_n(-x), \quad (10.21)$$

with $x > 0$. Consider the product of the functions $\omega_n(-x) \cdot \omega_n(x)$. We have

$$\omega_n(-x) \cdot \omega_n(x) = \left(1 - \frac{x^2}{n^2}\right)^n = \omega_n(\xi_n), \quad (10.22)$$

where

$$\xi_n = -\frac{x^2}{n}.$$

Since

$$\begin{aligned}\lim_{n \rightarrow \infty} \xi_n &= 0, \text{ we obtain} \\ \lim_{n \rightarrow \infty} \omega_n(\xi_n) &= 1.\end{aligned}$$

(See (10.20)). Now, from formula (10.22) we have:

$$\omega_n(-x) = \frac{\omega_n(\xi_n)}{\omega_n(x)},$$

and since both the numerator and the denominator in the right-hand side have definite limits as $n \rightarrow \infty$, it follows that:

$$\lim_{n \rightarrow \infty} \omega_n(-x) = \frac{\lim_{n \rightarrow \infty} \omega_n(\xi_n)}{\lim_{n \rightarrow \infty} \omega_n(x)} = \frac{1}{\exp(x)}.$$

Thus, we have shown that the function $\omega_n(-x)$, for positive x , converges to a definite limit, and hence we may define:

$$\lim_{n \rightarrow \infty} \omega_n(-x) = \exp(-x),$$

where

$$\exp(-x) = \frac{1}{\exp(x)}. \quad (10.23)$$

At the same time, we have also proven an important property of the exponential function:

$$\exp(-x) \cdot \exp(x) = 1, \quad \text{for } x > 0. \quad (10.24)$$

By renaming $-x$ as x , we obtain the general identity:

$$\exp(x) \cdot \exp(-x) = 1, \quad (10.25)$$

where x is now allowed to be negative. Also, note that when $x = 0$, we have $\omega_n(x) = 1$, and hence, by definition:

$$\exp(0) = 1. \quad (10.26)$$

Thus, for all values of x , we obtain the very important relation:

$$\exp(x) \cdot \exp(-x) = 1. \quad (10.27)$$

Therefore, we have proven that for any value of x , the quantity $\omega_n(x)$ tends to a definite limit as $n \rightarrow \infty$, so we may define:

$$\exp(x) = \lim_{n \rightarrow \infty} \omega_n(x). \quad (10.28)$$

From formula (10.18), it follows that for $|x| \leq 1$, we have:

$$\exp(x) = 1 + x + r(x), \text{ where } |r(x)| < x^2. \quad (10.29)$$

Fundamental Property of the Function $\exp(x)$

It turns out that the function $\exp(x)$ has the following most important property. If x and y are two real numbers, then we have the identity:

$$\exp(x) \exp(y) = \exp(x + y). \quad (10.30)$$

To prove this crucial equality, consider the product of the functions $\omega_n(x)$ and $\omega_n(y)$. We obtain:

$$\omega_n(x) \omega_n(y) = \left(1 + \frac{x}{n}\right)^n \left(1 + \frac{y}{n}\right)^n = \left(1 + \frac{x+y}{n} + \frac{xy}{n^2}\right)^n. \quad (10.31)$$

We have,

$$\begin{aligned} 1 + \frac{x+y}{n} + \frac{xy}{n^2} &= \left(1 + \frac{x+y}{n}\right) \cdot \left(1 + \frac{xy}{n(n+x+y)}\right) \\ &= \left(1 + \frac{x+y}{n}\right) \cdot \left(1 + \frac{\tilde{\xi}_n}{n}\right), \end{aligned} \quad (10.32)$$

where

$$\tilde{\xi}_n = \frac{xy}{n+x+y}. \quad (10.33)$$

From equations (10.31) and (10.32), we get:

$$\omega_n(x) \omega_n(y) = \omega_n(x+y) \omega_n(\tilde{\xi}_n). \quad (10.34)$$

Since $\lim_{n \rightarrow \infty} \tilde{\xi}_n = 0$, it follows that $\lim_{n \rightarrow \infty} \omega_n(\tilde{\xi}_n) = 1$ (see (10.20)), and therefore, taking the limit ($n \rightarrow \infty$) in (10.34) gives:

$$\exp(x) \exp(y) = \exp(x+y). \quad (10.35)$$

Thus, identity (10.30) is proven. This relation, proved for two multiplicands, is clearly also valid for any number of multiplicands. Taking all multiplicands to be

equal, we get:

$$(\exp(x))^p = \exp(px), \quad (10.36)$$

where p is a natural number. From this identity, and from identities (10.26) and (10.27), we obtain:

$$(\exp(x))^p = \exp(px), \quad (10.37)$$

where p is an arbitrary integer. Replacing the integer p with an integer q , and expressing $px = qx$, we obtain:

$$\left(\exp\left(\frac{y}{q}\right)\right)^q = \exp(y), \quad (10.38)$$

from which it follows that:

$$\exp\left(\frac{y}{q}\right) = (\exp(y))^{\frac{1}{q}}, \quad (10.39)$$

Raising both sides of the last relation to the power p , we get:

$$\exp\left(\frac{y}{q}\right)^p = (\exp(y))^{\frac{p}{q}} \quad (10.40)$$

In view of identity (10.36), the left-hand side can be written as:

$$\exp\left(\frac{y}{q}\right)^p = \exp\left(y\frac{p}{q}\right) \quad (10.41)$$

and therefore, we finally obtain:

$$\exp\left(\frac{p}{q}y\right) = (\exp(y))^{\frac{p}{q}} \quad (10.42)$$

Replacing y by x in this identity and setting $r = p/q$, we obtain the final formula:

$$\exp(rx) = (\exp(x))^r, \quad (10.43)$$

where r is an arbitrary rational number.

The Number e and the Function $\exp x$

By definition, we shall consider the number e to be given by the equation

$$e = \exp(1). \quad (10.44)$$

Thus, we have

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n. \quad (10.45)$$

This is the standard, commonly accepted definition of the number e . Substituting $x = 1$ into equation (10.43), we obtain

$$\exp(r) = e^r, \quad (10.46)$$

where r is any rational number. Hence, we have established that for any rational number r , the function $\exp(r)$ is nothing other than e^r , where exponentiation with a rational exponent r is understood in the standard algebraic sense.

For any real number x , not necessarily rational, the equality

$$e^x = \exp(x) \quad (10.47)$$

serves as the definition of the function e^x . The right-hand side of equation (10.47) has already been defined by us. The function e^x , initially defined in purely algebraic terms for rational values of x , is extended by this formula so that it makes sense for any real number x , not necessarily rational.

Such a definition of the function e^x for real values of x is the only reasonable one. The function resulting from this construction possesses good properties. These properties are as follows:

$$e^0 = 1, \quad e^{x+y} = e^x e^y. \quad (10.48)$$

These properties of the function e^x follow from properties (10.26) and (10.30) of the function $\exp(x)$, with which the function e^x coincides.

In addition, the function e^x has a derivative. Let us now proceed to compute it.

The Derivative of the Function e^x

To define the derivative of e^x , we use formula (1.21), setting

$$\xi = x + h. \quad (10.49)$$

Thus, we have

$$(e^x)' = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h}. \quad (10.50)$$

Since h is a small quantity, we may use formula (10.29) to approximate the function e^h , obtaining

$$e^h = 1 + h + r(h),$$

where

$$|r(h)| < h^2. \quad (10.51)$$

It follows that

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1. \quad (10.52)$$

Substituting this into formula (10.50), we get

$$(e^x)' = e^x. \quad (10.53)$$

Thus, the derivative of the function e^x is computed. It possesses the remarkable property of coinciding with

the function e^x itself. The same property is possessed by the function ce^x , where c is a constant. Namely,

$$(ce^x)' = ce^x. \quad (10.54)$$

It turns out that every function $f(x)$ satisfying the differential equation

$$f'(x) = f(x), \quad (10.55)$$

can be written in the form

$$f(x) = ce^x. \quad (10.56)$$

The proof is given in Exercise 3 to Section 11.

11 The Function $\ln x$

Consider the equation

$$e^y = x. \quad (11.1)$$

The solution of this equation with respect to y for a given value of x is called the *natural logarithm* of x , and it is denoted by

$$y = \ln x. \quad (11.2)$$

The function e^y in the variable y is defined for all values of y in the interval $-\infty < y < +\infty$, and it varies over the interval $0 < x < \infty$. Since the derivative of the function e^y is positive for all values of y , the solution $y = \ln x$ of equation (11.1) is defined and differentiable for all positive values of x (see Section 5, inverse function).

We now take the derivative of both sides of equation (11.1) with respect to x , considering $y = \varphi(x)$, and take the derivative of the left-hand side as a composite function (see formula (5.16)). Then, we obtain by this formula

$$(e^y)' = \varphi'(x) = 1. \quad (11.3)$$

In the expression $(e^y)'$, we must substitute $y = \varphi(x)$. Since

$$(e^y)' = e^y = e^{\varphi(x)} = x,$$

we obtain for $\varphi'(x)$ the expression

$$\varphi'(x) = \frac{1}{x}. \quad (11.4)$$

Here, $\varphi(x)$ is the natural logarithm of x . Thus, the derivative of $\ln x$ is given by the formula

$$(\ln x)' = \frac{1}{x}. \quad (11.5)$$

12 Expansion of the Function e^x into a Series

The formulas obtained in Section 10 make it possible to expand the function e^x into a power series.

From formula (10.6) it follows that

$$\lim_{n \rightarrow \infty} \gamma_n(k) = 1. \quad (12.1)$$

Now choose p so that $p + 1 > 10|x|$. Then from formula (10.12) we obtain

$$\frac{\gamma_n(k)}{k!} |x|^k \leq \frac{|x|^p}{p!} \cdot \frac{1}{10^{k-p}}.$$

Taking the limit as $n \rightarrow \infty$, we get

$$\frac{|x|^k}{k!} \leq \frac{|x|^p}{p!} \cdot \frac{1}{10^{k-p}}, \quad (12.2)$$

and since x is fixed, p is also fixed. Therefore, from formula (12.2), it follows that as $k \rightarrow \infty$, the quantity $|x|^k/k!$ tends to zero. Thus, for fixed x , we can find a sufficiently large value q such that

$$\frac{|x|^k}{k!} < 1 \quad \text{for } k \geq q. \quad (12.3)$$

Although we used a specific choice of the natural number p in the proof of this relation, the final result in (12.3) does not depend on this choice.

Let us now assume that in formula (10.16), $p > q$. Then from it we obtain

$$|r_n(p, x)| < \frac{|x|^p}{p!} \cdot \frac{|x|}{p+1-|x|} < \frac{|x|}{p+1-|x|} \quad (12.4)$$

From formula (10.13) it now follows that

$$|\omega_n(x) - s_n(p, x)| < \frac{|x|}{p+1-|x|}. \quad (12.5)$$

From formulas (10.14) and (10.1), it follows that in the limit as $n \rightarrow \infty$, the right-hand side of inequality (12.5) tends to zero. Therefore, this inequality implies that e^x expands into an infinite series:

$$\left| e^x - \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^p}{p!} \right) \right| < \frac{|x|}{p+1-|x|}.$$

12 Expansion of the Function e^x into a Series

Since the right-hand side of this inequality tends to zero with p increasing indefinitely, this inequality means that e^x decomposes into an infinite series.

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^p}{p!} + \dots \quad (12.6)$$

When $x = 1$, we obtain

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \quad (12.7)$$

Formulas (12.7) and (12.6) are convenient for calculating the value of e and the function e^x .

13 Epilogue: On the Theory of Limits

Personal experience convinces me that the initial acquaintance with mathematical analysis should not begin with the study of the theory of limits. While still a schoolboy, I had a fairly good command of the basics of differential and integral calculus and could use them to solve problems, though I was not even aware of the existence of the theory of limits. Only in my first year at university did I learn that such a theory exists, and I was quite surprised by it.

Historically, integral and differential calculus had already become well-developed areas of mathematics before the theory of limits appeared. The latter emerged as a sort of superstructure atop the already existing

theory.

Many physicists believe that the so-called rigorous definition of derivatives and integrals is not at all necessary for a good understanding of differential and integral calculus. I share this view. I think that beginning the presentation of mathematical analysis at school with the theory of limits may cause one to become completely bogged down in it, never reaching any substantial results. If acquaintance with the theory of limits is needed at all, it should come only after one has already become familiar with the substantive results of analysis. For this reason, I provide a very informal and highly intuitive description of the theory of limits only in this epilogue.

The Theory of Limits

The concept of a limit is always connected with the study of the behaviour of some function $f(\xi)$ in relation to the behaviour of its argument ξ . The relation between the behaviour of the function $f(\xi)$ and that of the argument ξ is of a very specific character. The question posed is: how does the quantity $f(\xi)$ behave when the variable quantity ξ approaches some fixed quantity x indefinitely?

If, in the process of ξ approaching the fixed quantity x , the variable quantity $f(\xi)$ also approaches some fixed quantity f_0 (we do not assume that the function $f(\xi)$ is defined at $\xi = x$), then the function $f(\xi)$ is said to tend to the limit f_0 as $\xi \rightarrow x$. This is written in the form:

$$\lim_{\xi \rightarrow x} f(\xi) = f_0. \quad (13.1)$$

In order not to give formal definitions, but only to provide an intuitive picture, we must imagine that ξ is an approximate value of the quantity x , and the accuracy of this approximation increases as ξ gets closer to x . In the same way, the quantity $f(\xi)$ should be regarded as an approximate value of the quantity f_0 , with the accuracy of this approximation increasing as the accuracy with which ξ approaches x increases.

Continuing with such an intuitive description, it becomes easy to understand the basic rules of taking limits. Suppose we have two functions $f(\xi)$ and $g(\xi)$, for which the following conditions hold:

$$\lim_{\xi \rightarrow x} f(\xi) = f_0, \quad \lim_{\xi \rightarrow x} g(\xi) = g_0, \quad (13.2)$$

i.e., the quantities $f(\xi)$ and $g(\xi)$ are increasingly accurate approximations of f_0 and g_0 respectively, as ξ becomes a more accurate approximation of x . Then it is clear that an approximate value of the sum $f_0 + g_0$ is

given by the sum $f(\xi) + g(\xi)$, and the accuracy of this approximation increases as the accuracies of approximations of f_0 by $f(\xi)$ and of g_0 by $g(\xi)$ improve. From this follows the first rule of limit theory:

$$\lim_{\xi \rightarrow x} [f(\xi) + g(\xi)] = f_0 + g_0 = \lim_{\xi \rightarrow x} f(\xi) + \lim_{\xi \rightarrow x} g(\xi). \quad (13.3)$$

The same applies to the product. Clearly, the product $f(\xi) \cdot g(\xi)$ is an approximate value of the product $f_0 \cdot g_0$. The accuracy of this approximation increases with the accuracy of the approximation of f_0 by $f(\xi)$ and of g_0 by $g(\xi)$. Hence follows the second rule of the theory of limits:

$$\lim_{\xi \rightarrow x} [f(\xi) \cdot g(\xi)] = f_0 \cdot g_0 = \lim_{\xi \rightarrow x} f(\xi) \cdot \lim_{\xi \rightarrow x} g(\xi). \quad (13.4)$$

Based on the very same reasoning, we also obtain the third rule of limit theory:

$$\lim_{\xi \rightarrow x} \frac{f(\xi)}{g(\xi)} = \frac{f_0}{g_0} = \frac{\lim_{\xi \rightarrow x} f(\xi)}{\lim_{\xi \rightarrow x} g(\xi)}. \quad (13.5)$$

This formula is valid, however, only under the condition that $g_0 \neq 0$. The next and final rule of the theory of limits concerns inequalities. If the quantity $f(\xi)$, which gives increasingly accurate approximations of the quantity f_0 , always remains less than or equal to the quantity $g(\xi)$, which gives increasingly accurate

approximations of the quantity g_0 , then it is clear that $f_0 \leq g_0$. In other words, from the relation

$$f(\xi) \leq g(\xi) \quad (13.6)$$

it follows that

$$f_0 \leq g_0, \quad \text{or} \quad \lim_{\xi \rightarrow x} f(\xi) \leq \lim_{\xi \rightarrow x} g(\xi). \quad (13.7)$$

A somewhat different but very similar variant of the theory of limits arises when the argument of a function is *not* a variable number ξ , approaching a constant number x , but a non-negative integer n , increasing without bound. In this case, we deal with functions $f(n)$ and $g(n)$, which are usually denoted slightly differently:

$$f(n) = f_n, \quad g(n) = g_n. \quad (13.8)$$

Here the question is: how does the value f_n behave as $n \rightarrow \infty$? If it turns out that f_n tends without bound towards a number f_0 , we write

$$\lim_{n \rightarrow \infty} f_n = f_0. \quad (13.9)$$

If, alongside this relation, an analogous relation holds for the second function:

$$\lim_{n \rightarrow \infty} g_n = g_0, \quad (13.10)$$

then based on the same intuitive considerations used in examining functions $f(\xi)$ and $g(\xi)$, we now obtain the main rules of the theory of limits in this context. They are as follows:

$$\lim_{n \rightarrow \infty} (f_n + g_n) = \lim_{n \rightarrow \infty} f_n + \lim_{n \rightarrow \infty} g_n, \quad (13.11)$$

$$\lim_{n \rightarrow \infty} (f_n \cdot g_n) = \lim_{n \rightarrow \infty} f_n \cdot \lim_{n \rightarrow \infty} g_n, \quad (13.12)$$

$$\lim_{n \rightarrow \infty} \frac{f_n}{g_n} = \frac{\lim_{n \rightarrow \infty} f_n}{\lim_{n \rightarrow \infty} g_n}. \quad (13.13)$$

Formula (13.13) is valid only if $\lim_{n \rightarrow \infty} g_n \neq 0$.

From the relation $f_n \leq g_n$, it follows that

$$\lim_{n \rightarrow \infty} f_n \leq \lim_{n \rightarrow \infty} g_n.$$

Continuous Functions

The concept of the limit is used in order to define the concept of a continuous function. Namely, a function $f(\xi)$, which is also defined at the value $\xi = x$, is said to be continuous at the value $\xi = x$ if the following relation holds:

$$\lim_{\xi \rightarrow x} f(\xi) = f(x).$$

A function $f(\xi)$ is said to be simply continuous if it is continuous at every value x of its argument.

If the function $g(\xi)$ is another continuous function, then from rules (13.1), (13.2), and (13.3) it follows that the sum of the functions $f(\xi) + g(\xi)$ is continuous, the product of the functions $f(\xi) \cdot g(\xi)$ is continuous, and the quotient of the functions is continuous except at those values for which $g(x) = 0$.

All functions considered in this book are, of course, continuous and, naturally, have derivatives; but I consider it unnecessary to state this at every step. This should be assumed, or better yet, intuitively felt.

14 Exercises

Section 1

Let us first point out some techniques for computing the derivative.

According to §1, the derivative $f'(x)$ of the function $f(x)$ is defined by the relation

$$f'(x) = \lim_{\xi \rightarrow x} \frac{f(\xi) - f(x)}{\xi - x}. \quad (14.1)$$

Thus, to find the derivative of the function $f(x)$, one must consider the quotient

$$\frac{f(\xi) - f(x)}{\xi - x} \quad (14.2)$$

and examine how this quotient behaves when x remains fixed while ξ approaches x without bound. If the fraction (14.2) approaches a certain definite limit, then this limit is denoted by $f'(x)$ and is the derivative of the function $f(x)$.

The limit of the quotient (14.2) is most easily computed in the case where the numerator $f(\xi) - f(x)$ can be directly divided by $\xi - x$. In this case, to evaluate the limit, it is sufficient to substitute $\xi = x$ into the resulting expression, and we obtain the derivative $f'(x)$. This is possible if in the expression $f(\xi) - f(x)$, one can factor out the binomial factor $\xi - x$. Since factoring out a binomial is not always a simple algebraic operation, it is often more convenient to change the notation so that one can factor out a monomial.

To do this, we set

$$h = \xi - x, \text{ or, equivalently,} \quad (14.3)$$

$$\xi = x + h. \quad (14.4)$$

Clearly, $h \rightarrow 0$ as $\xi \rightarrow x$. Thus, in the new notation, the definition (14.1) becomes

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (14.5)$$

Note that h is called the *increment of the independent*

variable, and the difference

$$f(x + \Delta) - f(x) \quad (14.6)$$

is the corresponding *increment of the function*. If in expression (14.6) we can factor out the factor h , then dividing by h becomes a purely algebraic operation, and therefore, to find the limit (14.5), it is sufficient to substitute $h = 0$ into the resulting quotient.

This technique works well when $f(x)$ is a rational function of x , but even in the presence of radicals, it may still work. For example, if the difference (14.6) can be written in the form $\sqrt{a} - \sqrt{b}$, where a and b are rational expressions, then by multiplying and dividing this difference by $\sqrt{a} + \sqrt{b}$, we obtain

$$\sqrt{a} - \sqrt{b} = \frac{a - b}{\sqrt{a} + \sqrt{b}}.$$

The expression $a - b$ no longer contains irrationality and can be divided either by $\xi - x$ or by Δ , depending on whether we are using definition (14.1) or (14.5) for the derivative.

Using these techniques, find the derivative of the function $f(x)$ given by the following different formulas:

$$1.1 \quad f(x) = ax^2 + bx + c.$$

$$\text{Answer: } f'(x) = 2ax + b.$$

14 Exercises

$$1.2 \quad f(x) = ax^3 + bx^2 + cx + d$$

$$\text{Answer: } f'(x) = 3ax^2 + 2bx + c$$

$$1.3 \quad f(x) = \frac{1}{x}$$

$$\text{Answer: } f'(x) = -\frac{1}{x^2}$$

$$1.4 \quad f(x) = \frac{1}{x^2}$$

$$\text{Answer: } f'(x) = -\frac{2}{x^3}$$

$$1.5 \quad f(x) = \frac{x-a}{x-b}$$

$$\text{Answer: } f'(x) = \frac{a-b}{(x-b)^2}$$

$$1.6 \quad f(x) = \frac{1}{ax^2 + bx + c}$$

$$\text{Answer: } f'(x) = \frac{-(2ax + b)}{(ax^2 + bx + c)^2}$$

$$1.7 \quad f(x) = \sqrt{x}$$

$$\text{Answer: } f'(x) = \frac{1}{2\sqrt{x}}$$

$$1.8 \quad f(x) = \sqrt{ax^2 + bx + c}$$

$$\text{Answer: } f'(x) = \frac{2ax + b}{2\sqrt{ax^2 + bx + c}}$$

Section 2

The rule for computing the derivative of a polynomial is so simple that it is impossible to provide any difficult or interesting exercises on it. Therefore, we shall give only two exercises here, the second of which will be used later. Let us compute the derivative of the following two polynomials.

$$2.1 \quad f(x) = x^4 + 4x^3 + 6x^2 + 4x + 1$$

$$\text{Answer: } f'(x) = 4(x^3 + 3x^2 + 3x + 1)$$

$$2.2 \quad f(x) = 3x^5 - 5(a^2 + b^2)x^3 + 15a^2b^2x$$

$$\text{Answer: } f'(x) = 15(x^4 - (a^2 + b^2)x^2 + a^2b^2)$$

Section 3

Consider the following polynomials $f(x)$. Determine the intervals on which the polynomial $f(x)$ is increasing and decreasing, and find all values of x at which the polynomial attains maxima and minima, distinguishing between them.

$$3.1 \quad f(x) = x^4 - px^2 + q$$

Answer: If $p \leq 0$, the polynomial $f(x)$ decreases on $-\infty < x \leq 0$ and increases on $0 \leq x < \infty$. It attains its minimum at $x = 0$. At this point, the second derivative is positive if $p < 0$, and zero if

$p = 0$. Thus, the condition (3.30) is not a necessary condition for a minimum of the function.

If $p > 0$, the polynomial $f(x)$ attains a maximum at $x = 0$ and two minima at $x = -\sqrt{p/2}$ and $x = \sqrt{p/2}$. The function decreases on $-\infty < x \leq -\sqrt{p/2}$, increases on $-\sqrt{p/2} \leq x \leq 0$, decreases again on $0 \leq x \leq \sqrt{p/2}$, and increases on $\sqrt{p/2} \leq x < \infty$.

- 3.2 $f(x) = 3x^5 - 5(a^2 + b^2)x^3 + 15a^2b^2x$ (see Exercise 2 for Section 2)

Answer: Let a and b be positive numbers, and assume $a > b$ for definiteness.

The polynomial $f(x)$ increases on $-\infty < x < -a$, decreases on $-a \leq x \leq -b$, increases on $-b \leq x \leq b$, decreases on $b \leq x \leq a$, and increases again on $a \leq x < \infty$. It attains maxima at $x = -a$ and $x = b$, and minima at $x = -b$ and $x = a$.

- 3.3 A rectangular sheet of iron with sides a and b , where $a \geq b$, is given. We will cut out squares with side $x < b/2$ at the corners of this sheet and bend each of the projections at a right angle upward from the resulting iron pattern. A box is thus formed with base dimensions $a - 2x$ and $b - 2x$, and height x , so the volume is $x(a - 2x)(b - 2x)$. Find the value of x at which the volume is maximal.

Answer: $x = (b - \sqrt{a^2 - ab + b^2})/6$ In the special case $a = b$, the solution simplifies to $x = a/6$.

Section 4

Sketch the graphs of the following polynomials $f(x)$.

4.1 $f(x) = x^4 - px^2 + q$ (see Exercise 1 of Section 3)

Answer: In Exercise 1 of Section 3, the polynomial $f(x)$ has already been sufficiently analysed. It remains only to determine under what conditions its graph intersects the x -axis, that is, when the equation $f(x) = 0$ has real roots, and how many such roots exist.

In the case $p \leq 0$, the equation $f(x) = 0$ has no real roots when $q > 0$, and it has two real roots when $q < 0$. If $p = 0$ and $q = 0$, then there is a quadruple root at $x = 0$; while if $p < 0$ and $q = 0$, there is a double root at $x = 0$.

When $p > 0$ and $q < 0$, there are two real roots. When $p > 0$, $q > 0$, the equation $f(x) = 0$ has four real roots if $p^2/4 - q > 0$, and no real roots if $p^2/4 - q < 0$. If $q > 0$ and $p^2/4 - q = 0$, then there are two double roots at $x = \pm\sqrt{p/2}$. If $p > 0$, $q = 0$, then there are three real roots, one of which is a double root.

$f(x) = 3x^5 - 5(a^2 + b^2)x^3 + 15a^2b^2x$ (see Exer-

cise 2 of Section 3)

Answer: In Exercise 2 of Section 3, the behaviour of the graph of the polynomial $f(x)$ has already been largely investigated.

It remains to clarify under what conditions the graph intersects the x -axis only at $x = 0$, or at more points, and to determine the number of intersection points.

The equation $f(x) = 0$ has:

§ five real roots if $a^2/b^2 > 5$,

§ one real root if $a^2/b^2 < 5$,

§ three real roots, two of which are double roots, if $a^2/b^2 = 5$.

4.2 Determine how many real roots the polynomial

$$y = g(x) = x^3 + a_1x^2 + a_2x + a_3, \quad (14.7)$$

has, depending on the coefficients a_1, a_2, a_3 .

Let us consider the equation:

$$f(x) = x^3 - px = c. \quad (14.8)$$

Clearly, when $p < 0$, this equation has only one real root. When $p = 0$, it also has only one real root, except in the case $c = 0$, in which case there is a triple root at $x = 0$. If $p > 0$, equation (14.8) has three real roots if:

$$f(x_2) \leq c < f(x_1), \quad (14.9)$$

where:

$$f(x_1) = \frac{2}{3}p\sqrt{\frac{p}{3}}, \quad x_2 = -\frac{2}{3}p\sqrt{\frac{p}{3}},$$

Then inequality (14.9) becomes:

$$|c| \leq \frac{2p}{3}\sqrt{\frac{p}{3}}. \quad (14.10)$$

For the boundary values of c in the interval (14.10), the equation has one simple root and one double root. Outside this interval, equation (14.10) has only one real root. Now, the polynomial (14.8) can be transformed into the form:

$$\eta = f(\xi) = \xi^3 - p\xi, \quad (14.11)$$

via the change of variables:

$$x = \xi + \alpha, \quad y = \eta + \beta \quad (14.12)$$

This transformation corresponds to a parallel shift of the coordinate system. Performing the substitution yields:

$$\begin{aligned} \eta = & \xi^3 + (3\alpha + a_1)\xi^2 + (3\alpha^2 + 2a_1\alpha + a_2)\xi \\ & + (\alpha^3 + a_1\alpha^2 + a_2\alpha + a_3 - \beta) \end{aligned} \quad (14.13)$$

In order for the polynomial (14.7) to be reduced to the form (14.11), it is sufficient that the following conditions are satisfied:

$$\begin{aligned} 3\alpha + a_1 &= 0, \\ \alpha^3 + a_1\alpha^2 + a_2\alpha + a_3 - \beta &= 0. \end{aligned} \quad (14.14)$$

From the first equation in (14.14), we find:

$$\alpha = -\frac{a_1}{3} \quad (14.15)$$

Substituting this value of α into the second equation of system (14.14), we obtain:

$$\beta = -\frac{a_1^3}{27} - \frac{a_1a_2}{3} + a_3 \quad (14.16)$$

Assuming that α and β are given by formulas (14.15) and (14.16), we can rewrite equation (14.13) in the form:

$$\eta = f(\xi) = \xi^3 - p\xi, \text{ where,} \quad (14.17)$$

$$p = \frac{a_1^2}{3} - a_2. \quad (14.18)$$

The geometric shape of the cubic parabola (14.7) coincides with that of the parabola (14.17), which depends only on the sign of p (14.18).

Thus, to determine the number of real roots of the polynomial $g(x)$ (see (14.7)), we observe that the polynomials $g(x)$ and $f(x)$ are related by the equation:

$$f(\xi) = g(x) - \beta.$$

Thus, the number of roots of the polynomial $g(x)$ is equal to the number of roots of the polynomial $f(\xi) + \beta$, or, equivalently, the number of roots of the equation $f(\xi) = -\beta$. But this question has already been resolved (see (14.8)). Therefore, for $p < 0$ (see (14.18)), the polynomial $g(x)$ has one real root, and for $p = 0$, the polynomial $g(x)$ also has one real root, except in the case when the value of β (see (14.16)) is zero.

Consider the case $p > 0$. If the value of β (see (14.16)) satisfies the condition

$$\beta \leq \frac{2}{3\sqrt{3}}p^{3/2}, \quad (14.19)$$

(see (14.10)), then the polynomial $g(x)$ has three real roots, with two roots merging into one at the ends of the interval (14.19). By substituting the values of β and p from the relations (14.16) and (14.18) into (14.19) and squaring the resulting

equation, we obtain the condition:

$$\left(\frac{2a_1^3}{27} - \frac{a_1a_2}{3} + a_3 \right)^2 \leq \frac{4}{27} \left(\frac{a_1^2}{3} - a_2 \right) \quad (14.20)$$

The obtained condition is a complete criterion for the existence of three real roots of the polynomial $g(x)$, since its satisfaction excludes the case $p < 0$. Indeed, the right-hand side of inequality (14.20) is negative when $p < 0$, while the left-hand side, being a square, cannot be negative.

Now, let us indicate the approach to studying the graph of the fourth-degree polynomial

$$y = h(x) = x^4 + b_1x^3 + b_2x^2 + b_3x + b_4.$$

To determine its behaviour, it is necessary to study the behaviour of the polynomial

$$h'(x) = 4x^3 + 3b_1x^2 + 2b_2x + b_3.$$

This polynomial is already of degree 3, and the question of how many real roots it has already been resolved.

Since the polynomial $h(x)$ increases without bound and remains positive as $x \rightarrow -\infty$ and $x \rightarrow \infty$, then in the case where the polynomial $h'(x)$ has only one real root, the polynomial $h(x)$ has only

one minimum. If, however, the polynomial $h'(x)$ has three distinct real roots, then the polynomial $h(x)$ has two minima and one maximum.

The shape of the graph of the function $h(x)$ in these two cases is shown in Figure 5 (a) and Figure 5 (b). In the case where two real roots merge into one, the graph has the appearance shown in Figure 5 (c).

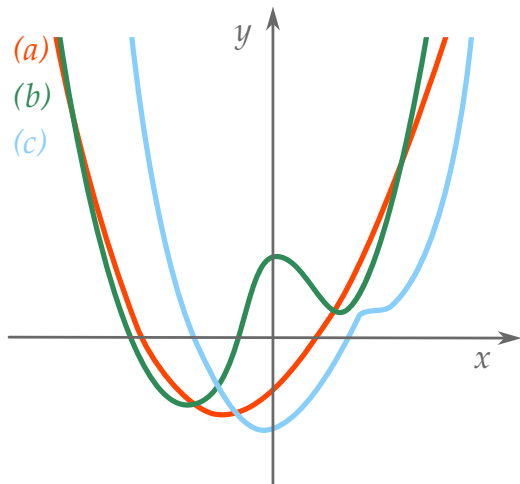


Figure 5: The shape of the graph of the function $h(x)$ for different cases.

4.3 Prove that the graph of the function $y = f(x)$ has

a point of inflection at $x = x_0$ if

$$f''(x_0) = 0, \quad f'''(x_0) \neq 0.$$

Here, $f'''(x)$ denotes the third derivative of $f(x)$, i.e., it is defined as

$$f'''(x) = (f''(x))'.$$

Section 5

5.1 Plot the graphs of the functions

$$y = \sin x,$$

$$y = \cos x.$$

Find the intervals on which each of the functions $\sin x$ and $\cos x$ is decreasing. Find the points of maximum and minimum of each function. Find the points where the graphs intersect the x -axis. Find the points of inflection of each graph.

5.2 Plot the graph of the function

$$y = \tan x.$$

Answer. Since the function $\tan x$ is periodic with period π , we have

$$\tan(x + k\pi) = \tan x,$$

where k is any integer. Therefore, to construct the complete graph of the function $\tan x$, it suffices to construct it over the interval $-\pi/2 < x < \pi/2$. Over this interval, the function $\tan x$ increases from $-\infty$ to $+\infty$, passing through zero at $x = 0$, where the graph has a point of inflection.

To construct the complete graph of $\tan x$, shift this section by $k\pi$ along the x -axis, where k is an arbitrary integer. Each such shift gives a distinct branch of the graph of $\tan x$, which does not intersect the others.

5.3 Plot the graph of the function

$$y = f(x) = \frac{a}{x}, \quad \text{where } a > 0.$$

Answer. The function $f(x)$ is not defined at $x = 0$ and therefore consists of two separate branches: one lying above the x -axis for $x > 0$ and one lying below the x -axis for $x < 0$. Since

$$f'(x) = \left(\frac{a}{x}\right)' = -\frac{a}{x^2} < 0,$$

the function $f(x)$ is decreasing on both of its branches.

Let us examine it more closely for $x > 0$. As x remains positive and approaches zero, the function $f(x)$ increases without bound, so the graph approaches the y -axis. Likewise, as $x \rightarrow \infty$, the

value of $f(x)$ tends to zero, so the graph approaches the x -axis. The relationship between x and y can also be written in the form

$$xy = a.$$

The curve defined by this equation is clearly symmetric with respect to the bisector of the first and third quadrants.

5.4 Plot the graph of the function

$$y = f(x) = x^2 \cos x.$$

Answer. First of all, it is clear that the function $f(x)$ is even, namely $f(-x) = f(x)$. Thus, the graph of this function is symmetric with respect to the y -axis, and therefore it suffices to consider the behaviour for $x \geq 0$.

For positive values of x , the sign of the function $x^2 \cos x$ coincides with the sign of $\cos x$. Hence, for $x > \pi/2$, the graph consists of waves similar to those of the function $\cos x$, but these waves grow in amplitude as x increases, since $\cos x$ is multiplied by x^2 .

Clearly, on each wave lying above the x -axis, the function $f(x)$ attains a maximum, and on each wave below the x -axis, the function $f(x)$ attains a minimum.

The interval $-\pi/2 \leq x < \pi/2$ requires closer examination. At the endpoints of this interval, the function $\cos x$ is zero, so the function $f(x)$ is also zero.

For intermediate values of x , the function $f(x)$ is non-negative. At $x = 0$, the graph passes through the origin, where $f(x)$ attains its minimum. On the interval $0 < x < \pi/2$, the function $f(x)$ is positive, except at the endpoints where it is zero, and therefore it must attain a maximum somewhere in this interval.

To find the values of x where $f(x)$ attains its maxima and minima, we must set its derivative equal to zero, i.e., solve the equation:

$$2x \cos x - x^2 \sin x = 0.$$

Dividing both sides of this equation by $x^2 \cos x$ (assuming $x \neq 0$ and $\cos x \neq 0$), we obtain:

$$\frac{2}{x} = \tan x.$$

Thus, to find the maximum and minimum points of $f(x)$, we must find the intersection points of the graphs of the functions

$$y = \frac{2}{x} \quad \text{and} \quad y = \tan x.$$

Since the graph of $y = \frac{2}{x}$ approaches the x -axis as x increases, for relatively large x the solution of the equation $2/x = \tan x$ is approximately the solution to $\tan x = 0$, i.e., approximately equal to $k\pi$, where k is an integer.

Geometrically, it is seen that the actual solution to the equation $2/x = \tan x$ is slightly greater than $k\pi$. This shows that on each wave of the graph of the function $f(x)$, the function attains either its maximum approximately where $\cos x$ has a maximum, or its minimum approximately where $\cos x$ has a minimum, with only a slight rightward shift.

- 5.5 Compute the derivatives of the functions given in the exercises for Section 1, using the rules from Section 5.

Compute the derivatives of the following functions.

5.6 $f(x) = (\sin x)^n$

Answer: $f'(x) = n(\sin x)^{n-1} \cos x$.

5.7 $f(x) = \sin(x^n)$

Answer: $f'(x) = \cos(x^n) \cdot nx^{n-1}$.

5.8 $f(x) = \sqrt[n]{x}$

Answer: $f'(x) = \frac{1}{n\sqrt[n]{x^{n-1}}}$.

- 5.9 Let us consider a function $\psi(y)$ of the variable y . Its derivative with respect to y is denoted by $\psi'(y)$. Suppose now that $y = ax$, where a is

a constant. Substituting $y = ax$ into $\psi(y)$, we obtain a function

$$f(x) = \psi(ax).$$

Compute the derivative $f'(x)$.

Answer:

$$f'(x) = \psi'(ax) \cdot a.$$

To compute $f'(x)$, we first find $\psi'(y)$, substitute $y = ax$, and multiply the result by a . Using Formula (5.16), we have:

$$f'(x) = \psi'(ax) \cdot (ax)' = \psi'(ax) \cdot a.$$

- 5.10 Let $\psi(y)$ be a function of the variable y . Its derivative is denoted $\psi'(y)$. Suppose $y = x + a$. Substituting this into $\psi(y)$ gives

$$f(x) = \psi(x + a).$$

Compute the derivative $f'(x)$.

Answer

$$f'(x) = \psi'(x + a).$$

From (5.16):

$$f'(x) = \psi'(x + a) \cdot (x + a)' = \psi'(x + a).$$

- 5.11 Consider the function

$$\psi(y) = \sin y. \quad (14.21)$$

on the interval $-\pi/2 \leq y \leq \pi/2$. We know that

$$\sin' y = \cos y.$$

and that $\cos y > 0$ for all interior points of the interval $[-\pi/2, \pi/2]$. Hence

$$\sin' y = \cos y > 0.$$

In our case:

$$\begin{aligned} b_1 &= -\frac{\pi}{2}, & b_2 &= \frac{\pi}{2}, \\ a_1 &= \sin\left(-\frac{\pi}{2}\right) = -1, & a_2 &= \sin\left(\frac{\pi}{2}\right) = +1. \end{aligned}$$

Thus, the solution $y = \varphi(x)$ of the equation (14.21) is defined for all x in the interval $-1 \leq x \leq 1$.

This solution is denoted by $y = \arcsin x$. Then we have

$$\sin(\arcsin x) = x.$$

and also

$$\arcsin(\sin y) = y. \quad (14.22)$$

The task is to use formula (5.21) to compute the derivative $\arcsin' x$ of the function $\arcsin x$.

Answer

$$\arcsin' x = \frac{1}{\sqrt{1-x^2}}. \quad (14.23)$$

According to (5.21):

$$\arcsin' x = \frac{1}{\psi'(y)} = \frac{1}{\cos y}$$

Since $x = \sin y$, we know that

$$\cos y = \sqrt{1 - x^2}.$$

Substituting into the previous expression, we get formula (14.23).

5.12 The function $\arccos x$ is defined by the condition

$$\cos(\arccos x) = x.$$

Compute the derivative of the function $\arccos'(x)$.

Answer:

$$\arccos' x = -\frac{1}{\sqrt{1 - x^2}}. \quad (14.24)$$

5.13 The function $\arctan x$ is defined by the condition

$$\tan(\arctan x) = x,$$

where only solutions with $|\arctan x| < \pi/2$ are considered.

Using formula (5.16), compute the derivative $\arctan'(x)$ of the function $\arctan x$.

Answer:

$$\arctan'(x) = \frac{1}{1 + x^2}. \quad (14.25)$$

5.14 Compute the derivative of the function

$$f(x) = \arcsin x + x\sqrt{1-x^2}. \quad (14.26)$$

Answer:

$$f'(x) = 2\sqrt{1-x^2}. \quad (14.27)$$

Section 6

6.1 Find the antiderivative $h(x)$ of the function $f(x) = x^3 - px$.

Answer:

$$h(x) = \frac{x^4}{4} - \frac{px^2}{2}.$$

6.2 Find the antiderivative $h(x)$ of the function $f(x) = \sqrt{r^2 - x^2}$.

Answer:

$$h(x) = \frac{r^2}{2} \arcsin \frac{x}{r} + \frac{x}{2} \sqrt{r^2 - x^2}. \quad (14.28)$$

To prove this, define

$$\psi(y) = \arcsin y + y\sqrt{1-y^2}.$$

Using formula (14.27), we have:

$$\psi'(y) = 2\sqrt{1-y^2}.$$

Let us now compute the derivative of the function $\psi(x/r)$. According to the formula from Exercise 9 Section 5, we have:

$$\left(\psi\left(\frac{x}{r}\right)\right)' = \frac{1}{r}\psi'\left(\frac{x}{r}\right).$$

Hence, the formula (14.28) is valid.

- 6.3 In applications of mathematics, *differential equations* often play an important role. In such equations, the unknowns are functions, and the equations involve not only the function itself but also its derivatives. One of the simplest and most important equations is the following:

$$f''(t) + \omega^2 f(t) = 0. \quad (14.29)$$

Here, t is the independent variable, and $f(t)$ is the unknown function.

Let us explain how this equation arises in mechanics. Consider the motion of a point along a straight line, which we take as the x -axis. Let x be the coordinate of the moving point. The motion of the point is described by the equation

$$x = f(t),$$

where t is time, and x is the coordinate of the point at time t .

Suppose the point has mass m and is subject to an attractive force directed towards the origin, proportional to the displacement from the origin. The force is given by $-kx$, where k is a positive constant. Then, according to the laws of mechanics:

$$mf''(t) = -kf(t).$$

which is equivalent to equation (14.29).

On the left is the mass multiplied by the acceleration of the point, and on the right is the magnitude of the acting force. Denoting the positive quantity k/m by ω^2 , we can rewrite the last equation in the form of equation (14.29). Our task is to find a solution to equation (14.29), or more precisely, all solutions of this equation, since there is more than one.

Answer. Any solution of equation (14.29) can be written in the form

$$f(t) = a \sin(\omega t + \alpha), \quad (14.30)$$

where a and α are arbitrary constants. Let us prove this. First, note that if $f_1(t)$ and $f_2(t)$ are two solutions of equation (14.29), that is, the identities

$$\begin{aligned} f_1''(t) + \omega^2 f_1(t) &= 0; \\ f_2''(t) + \omega^2 f_2(t) &= 0, \end{aligned} \quad (14.31)$$

hold, then the function,

$$f(t) = c_1 f_1(t) + c_2 f_2(t), \quad (14.32)$$

where c_1 and c_2 are constants, is also a solution of equation (14.29). Indeed, substituting the function (14.32) into equation (14.29), we obtain:

$$\begin{aligned} c_1 f_1''(t) + c_2 f_2''(t) + c_1 \omega^2 f_1(t) + c_2 \omega^2 f_2(t) \\ = c_1 (f_1''(t) + \omega^2 f_1(t)) + c_2 (f_2''(t) + \omega^2 f_2(t)) = 0, \end{aligned}$$

by virtue of (14.31).

Let us now show that the functions

$$\begin{aligned} f_1(t) &= \sin(\omega t); \\ f_2(t) &= \cos(\omega t), \end{aligned} \quad (14.33)$$

are solutions of equation (14.29). Indeed, substituting $f(t) = \sin(\omega t)$ into equation (14.29) gives:

$$\begin{aligned} (\sin'(\omega t))' + \omega^2 \sin(\omega t) \\ = (\omega \cos(\omega t))' + \omega^2 \sin(\omega t) \\ = -\omega^2 \sin(\omega t) + \omega^2 \sin(\omega t) = 0. \end{aligned}$$

Substituting $f(t) = \cos(\omega t)$ gives:

$$\begin{aligned} (\cos'(\omega t))' + \omega^2 \cos(\omega t) \\ = (-\omega \sin(\omega t))' + \omega^2 \cos(\omega t) \\ = -\omega^2 \cos(\omega t) + \omega^2 \cos(\omega t) = 0. \end{aligned}$$

Thus, by formula (14.32), the function

$$f(t) = c_1 \sin(\omega t) + c_2 \cos(\omega t) \quad (14.34)$$

is a solution of equation (14.29). It turns out that every solution $f(t)$ of equation (14.29) can be written in the form (14.34).

To prove this, let us take two arbitrary solutions $f_1(t)$ and $f_2(t)$ of equation (14.29) (see (14.31)), and construct the auxiliary function

$$g(t) = f_1(t)f_2'(t) - f_1'(t)f_2(t). \quad (14.35)$$

We shall prove that

$$g'(t) = 0. \quad (14.36)$$

Indeed, we have

$$\begin{aligned} g'(t) &= f_1'(t)f_2'(t) + f_1(t)f_2''(t) - f_1''(t)f_2(t) - f_1'(t)f_2'(t) \\ &= f_1(t)(f_2''(t) + \omega^2 f_2(t)) - f_2(t)(f_1''(t) + \omega^2 f_1(t)) \\ &= 0. \end{aligned}$$

Thus, (14.36) is valid and $g(t)$ is constant:

$$f_1(t)f_2'(t) - f_1'(t)f_2(t) = c. \quad (14.37)$$

Now let $f(t)$ be an arbitrary solution of equation (14.29). Substituting into identity (14.37) the

functions

$$\begin{aligned}f_1(t) &= f(t), \\f_2(t) &= \sin(\omega t),\end{aligned}$$

we get

$$f(t) \cdot \omega \cos(\omega t) - f'(t) \cdot \sin(\omega t) = c_2 \omega. \quad (14.38)$$

Substituting into identity (14.37) the functions

$$\begin{aligned}f_1(t) &= f(t), \\f_2(t) &= \cos(\omega t),\end{aligned}$$

we obtain

$$-f(t) \cdot \omega \sin(\omega t) - f'(t) \cdot \cos(\omega t) = -c_1 \omega. \quad (14.39)$$

Here, $-c_1 \omega$ and $c_2 \omega$ are arbitrary constants. Let us eliminate the function $f'(t)$ from equations (14.38) and (14.39). To do this, multiply (14.38) by $\cos(\omega t)$ and (14.39) by $\sin(\omega t)$, and subtract the second result from the first. We get:

$$f(t) \omega \cos^2(\omega t) + f(t) \omega \sin^2(\omega t) = c_2 \omega \cos(\omega t) + c_1 \omega \sin(\omega t),$$

from which equality (14.34) follows.

Now choose such constants a and α so that

$$\begin{aligned}c_1 &= a \cos \alpha; \\c_2 &= a \sin \alpha,\end{aligned}$$

which is always possible. Then $f(t)$ can be written in the form:

$$f(t) = a \cos(\alpha) \sin(\omega t) + a \sin(\alpha) \cos(\omega t) = a \sin(\omega t + \alpha).$$

Thus, it is proven that every solution of equation (14.29) is of the form (14.30).

The equation

$$x = a \sin(\omega t + \alpha), \quad (14.40)$$

describes the oscillatory motion of a point x with amplitude a and initial phase α . If we set $\omega = 2\pi/T$, then equation (14.40) becomes

$$x = a \sin\left(\frac{2\pi t}{T} + \alpha\right),$$

where T is the period of the oscillatory motion described by (14.40).

Section 7

7.1 Consider the cubic parabola L given by the equation

$$y = f(x) = x^3 - px,$$

where p is a positive number. The graph L of the function $f(x)$ intersects the x -axis at three points

a_{-1} , a_0 , and a_1 , whose abscissae are respectively $-\sqrt{p}$, 0, and $\sqrt{p}$.

Let L_1 be the arc of the curve L from point a_{-1} to point a_0 . This arc lies above the x -axis and together with the segment $[a_{-1}, a_0]$ of the x -axis bounds a region P in the plane. We are to compute the area S of region P .

Answer:

$$S = \frac{p^2}{4}.$$

From formula (7.16):

$$S = \int_{-\sqrt{p}}^0 f(t) dt. \quad (14.41)$$

The function

$$h(x) = \frac{1}{4}x^4 - \frac{p}{2}x^2.$$

is an antiderivative of $f(x)$, so the definite integral (14.41) can be written as:

$$\int_{-\sqrt{p}}^0 f(t) dt = h(0) - h(-\sqrt{p}) = \frac{p^2}{4}.$$

Hence

$$S = \frac{p^2}{4}.$$

7.2 Compute the area S of a circle of radius r using formula (7.16).

Answer: $S = \pi r^2$.

To compute this, consider a circle of radius r with its centre at the origin. The equation is: $x^2 + y^2 = r^2$ or $y = \pm\sqrt{r^2 - x^2}$.

We consider the portion of the circle lying in the first quadrant. Its equation becomes: $y = f(x) = \sqrt{r^2 - x^2}$, for x between $[0, r]$. Thus, the area of this portion is given by:

$$S = \int_0^r \sqrt{r^2 - t^2} \, dt.$$

The function

$$h(x) = \frac{r^2}{2} \arcsin\left(\frac{x}{r}\right) + \frac{x}{2} \sqrt{r^2 - x^2},$$

is an antiderivative of $f(x)$ (see Exercise 2 to Section 6). Therefore:

$$\int_0^r \sqrt{r^2 - t^2} \, dt = h(r) - h(0) = \frac{r^2}{2} \cdot \frac{\pi}{2} = \frac{\pi r^2}{4}.$$

Hence, the area of the full circle is:

$$S = 4 \cdot \frac{\pi r^2}{4} = \pi r^2.$$

Section 8

8.1 Prove that the series

$$S = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)} + \cdots \quad (14.42)$$

consisting of an infinite number of positive terms, converges, and compute its sum.

Answer: $S = 1$.

To compute the sum S of the infinite series (14.42), let us consider the partial sum:

$$S_n = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots + \frac{1}{n(n+1)}. \quad (14.43)$$

Note the identity:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Using this identity, we can rewrite the partial sum as a telescoping series:

$$\begin{aligned} S_n &= \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

As $n \rightarrow \infty$, we have $\lim_{n \rightarrow \infty} S_n = 1$. Therefore, the sum of the infinite series (14.42) is $S = 1$.

Section 10

10.1 Prove that the function

$$f(x) = \frac{e^x}{x^k},$$

where k is an arbitrary natural number, always tends to infinity as $x \rightarrow \infty$.

Answer. From the result established in Section 10, we have the identity:

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n.$$

From this it follows

$$e^x > \gamma_n(k+1) \frac{x^{(k+1)}}{(k+1)!}.$$

(see formula (10.7)). Passing to the limit as $n \rightarrow \infty$, and considering only positive values of x , we have:

$$e^x \geq \frac{x^{k+1}}{(k+1)!}.$$

Thus, for positive values of x , we have

$$f(x) = \frac{e^x}{x^k} > \frac{x^{k+1}}{(k+1)! x^k} = \frac{x}{(k+1)!}.$$

Clearly, as $x \rightarrow \infty$, the right-hand side of the last inequality tends to infinity. Hence, the statement is proved.

Section 11

11.1 Prove that the function

$$f(x) = \frac{(\ln x)^k}{x},$$

tends to zero as $x \rightarrow \infty$.

Answer. Use Exercise 1 from Section 10 for the proof.

11.2 Sketch the graph of the function

$$y = f(x) = \frac{\ln x}{x}.$$

Answer. The function $f(x)$ on the interval $0 < x \leq e$ increases, ranging from $-\infty < f(x) \leq \frac{1}{e}$, and then on the interval $e \leq x \leq \infty$ it decreases, tending towards zero. At the point $x = e$, it reaches a maximum.

11.3 Solve the differential equation

$$f'(x) = \lambda f(x), \quad (14.44)$$

where $\lambda \neq 0$ is a given number.

Answer.

$$f(x) = ce^{\lambda x}, \quad (14.45)$$

where c is an arbitrary constant.

First of all, it is clear that the function (14.45) is a solution of equation (14.44), which can be verified by direct substitution. Now we shall prove that every solution $f(x)$ of equation (14.44) has the form (14.45). To this end, we introduce the auxiliary function

$$g(x) = f(x)e^{-\lambda x}$$

and prove that

$$g'(x) = 0.$$

Indeed, we have:

$$\begin{aligned} g'(x) &= f'(x)e^{-\lambda x} - \lambda f(x)e^{-\lambda x} \\ &= e^{-\lambda x}(f'(x) - \lambda f(x)) \\ &= 0. \end{aligned}$$

Thus, the function $g(x)$ is constant, say c , and hence

$$f(x)e^{-\lambda x} = c,$$

which implies (14.45).

Equation (14.44) describes, in particular, radioactive decay, where the quantity of the undecayed substance at time t is denoted by $f(t)$. The quantity of substance that decays in the interval from t to τ is proportional to the amount of undecayed

substance and the time interval $\tau - t$. Thus we have:

$$f(\tau) - f(t) = -\alpha f(t)(\tau - t).$$

Dividing this relation by $\tau - t$ and passing to the limit gives:

$$f'(t) = -\alpha f(t).$$

According to the result just proved, the solution of this equation is:

$$f(t) = c e^{-\alpha t}. \quad (14.46)$$

If at the initial time $t = 0$ the amount of undecayed substance is f_0 , then substituting into (14.46) gives:

$$f(0) = c e^{-\alpha \cdot 0} = c.$$

Thus, the amount of undecayed substance is given by the formula:

$$f(t) = f_0 e^{-\alpha t}.$$

The time during which half the initial substance decays is called the *half-life*, denoted t_0 . From (11.3), this is found by solving:

$$e^{-\alpha t_0} = \frac{1}{2},$$

which yields:

$$t_0 = \frac{\ln 2}{\alpha}.$$

- 11.4 It is easy to verify that $\ln 1 = 0$. From this and from formula (11.5), it follows that

$$\ln x = \int_1^x \frac{d\xi}{\xi}.$$

Using this formula as the definition of the logarithmic function $\ln x$, prove the fundamental property:

$$\ln(x_1 x_2) = \ln x_1 + \ln x_2.$$

Also, prove that the function inverse to $\ln x$, that is, the solution $y = \varphi(x)$ of the equation

$$\ln y = x,$$

is the function $y = e^x$.

Section 13

- 13.1 Prove that the function $\sin(1/x)$ does not tend to any limit as $x \rightarrow 0$. More precisely, it is possible to choose a sequence of positive numbers $x_1, x_2 \dots x_k \dots$ such that

$$\lim_{k \rightarrow \infty} \sin\left(\frac{1}{x_k}\right) = c,$$

where $c \in (-1, 1)$ is arbitrary.

Answer. Let us choose the sequence

$$x_k = \frac{1}{2\pi k + \alpha}, \quad \alpha \geq 0.$$

Then clearly

$$\begin{aligned} \lim_{k \rightarrow \infty} x_k &= 0, \\ \lim_{k \rightarrow \infty} \sin\left(\frac{1}{x_k}\right) &= \sin \alpha. \end{aligned}$$

13.2 Show that the function

$$f(x) = x \sin\left(\frac{1}{x}\right) \quad (14.47)$$

is continuous at $x = 0$, but not differentiable at that point.

Answer. Consider the difference quotient to find $f'(0)$:

$$\frac{f(\xi) - f(0)}{\xi - 0} = \frac{\xi \sin\left(\frac{1}{\xi}\right)}{\xi} = \sin\left(\frac{1}{\xi}\right). \quad (14.48)$$

As $x \rightarrow 0$, the function $\sin(1/x)$ does not approach any finite limit (see Exercise 1), and hence this quotient (14.48) has no limit, and therefore the function (14.47) does not have a derivative at $x = 0$.